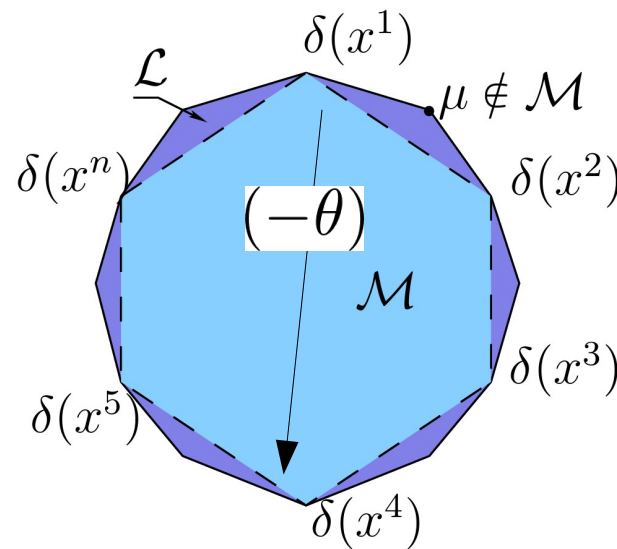


When the energy minimization is solvable/solved exactly?

- acyclic graphs: local polytope = marginal polytope

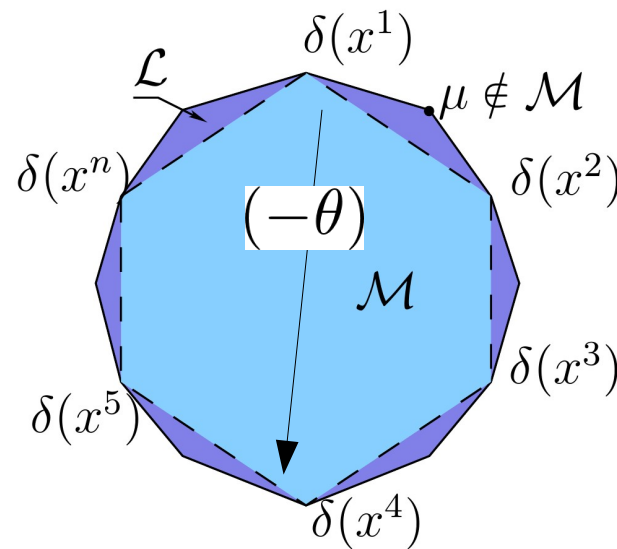
When the energy minimization is solvable/solved exactly?

- acyclic graphs: local polytope = marginal polytope
- general graphs, when $\langle \theta, \delta(x) \rangle$ is minimized by an integer vertex of the local polytope



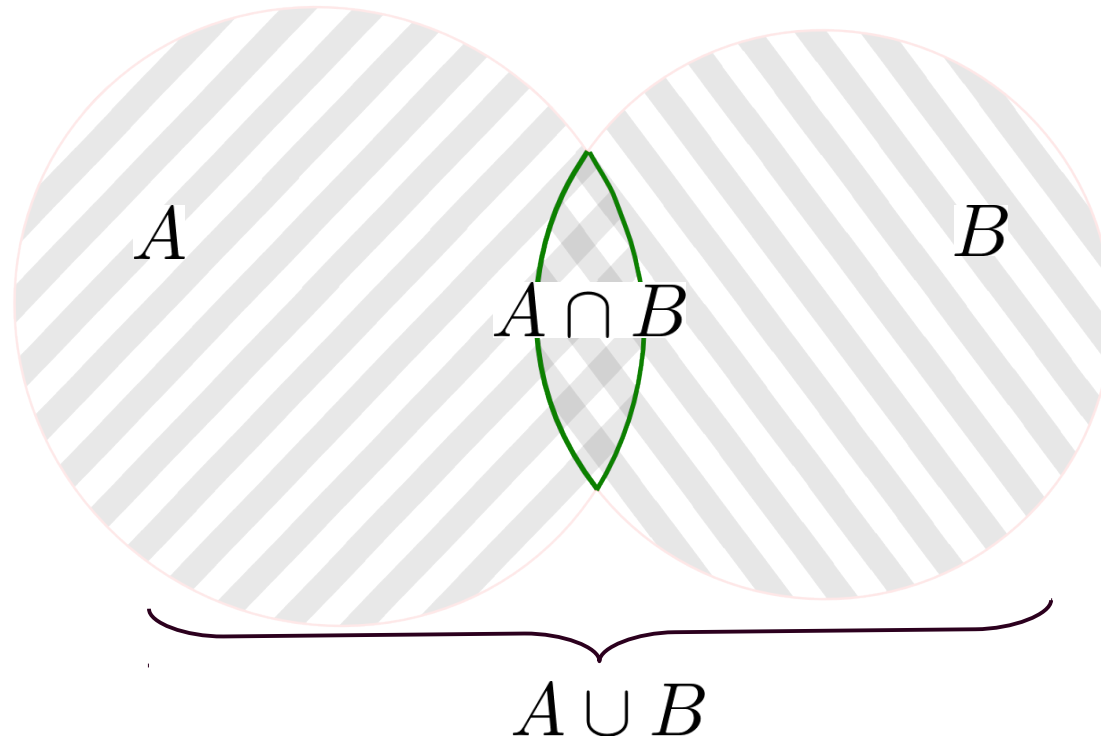
When the energy minimization is solvable/solved exactly?

- acyclic graphs: local polytope = marginal polytope
- general graphs, when $\langle \theta, \delta(x) \rangle$ is minimized by an integer vertex of the local polytope



Are there such potentials, which always have this property?

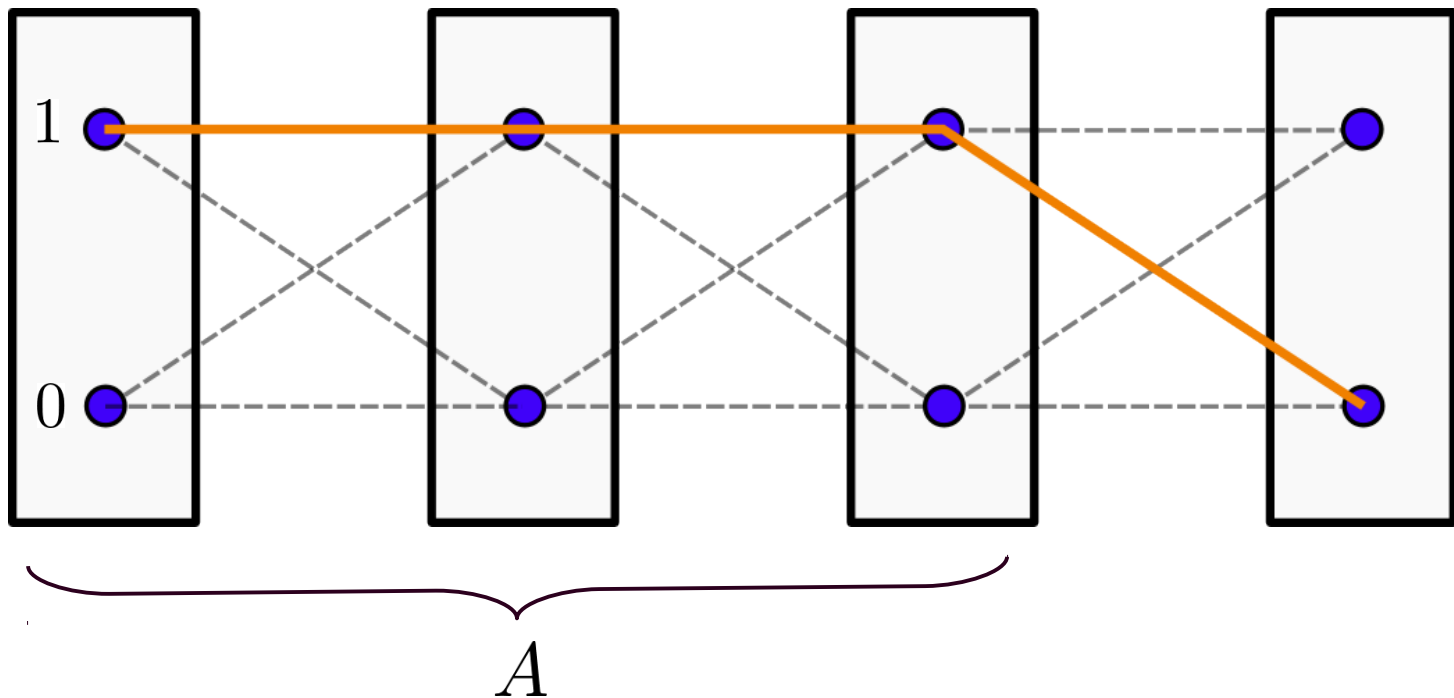
$$E(A) + E(B) \geq E(A \cap B) + E(A \cup B)$$



$\min_A E(A)$ – submodular minimization, polynomially solvable

$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

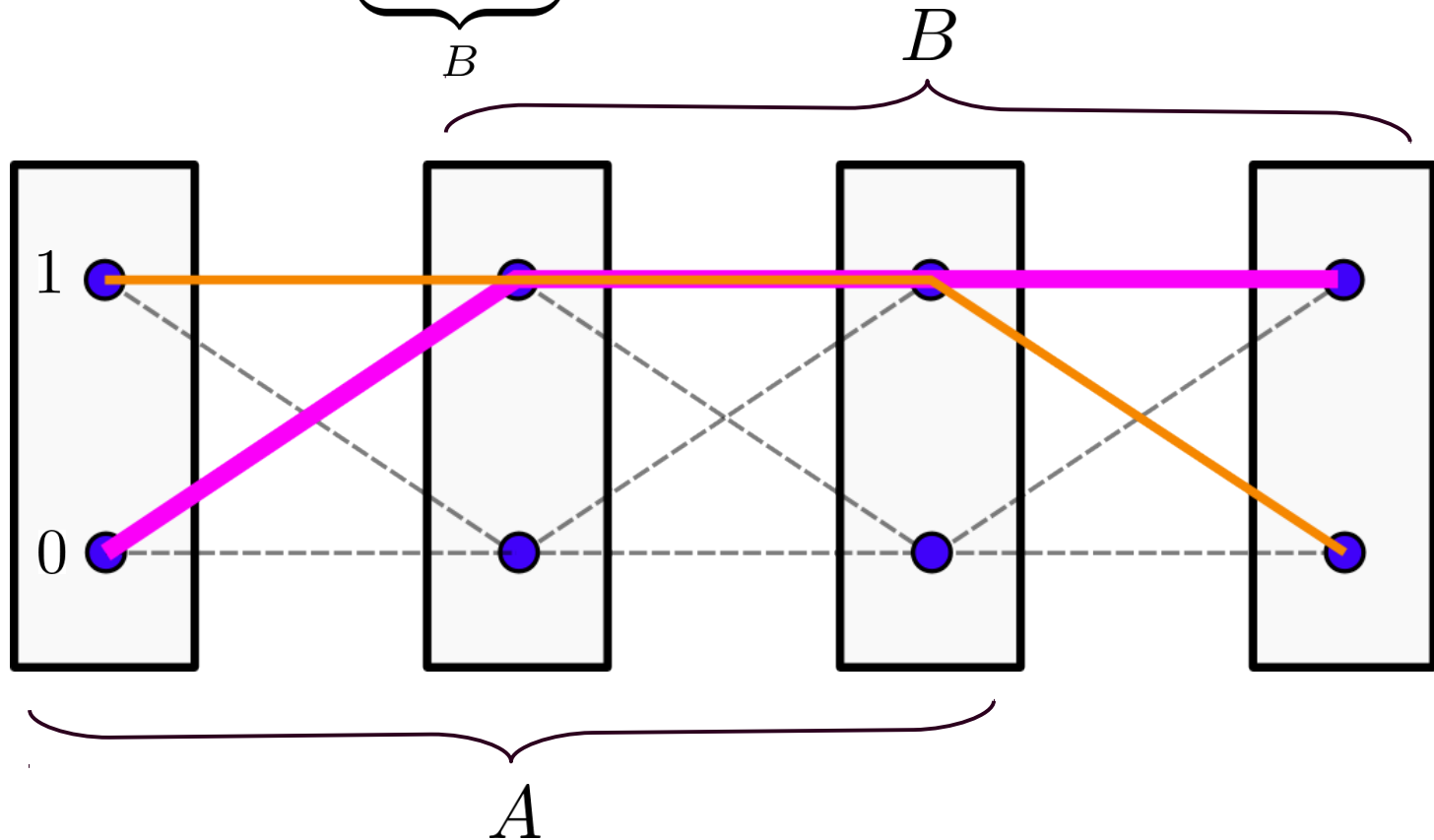
$$A \Leftrightarrow x^A = \{\underbrace{1, \dots, 1}_A, 0, \dots, 0\}$$



Submodular Energies

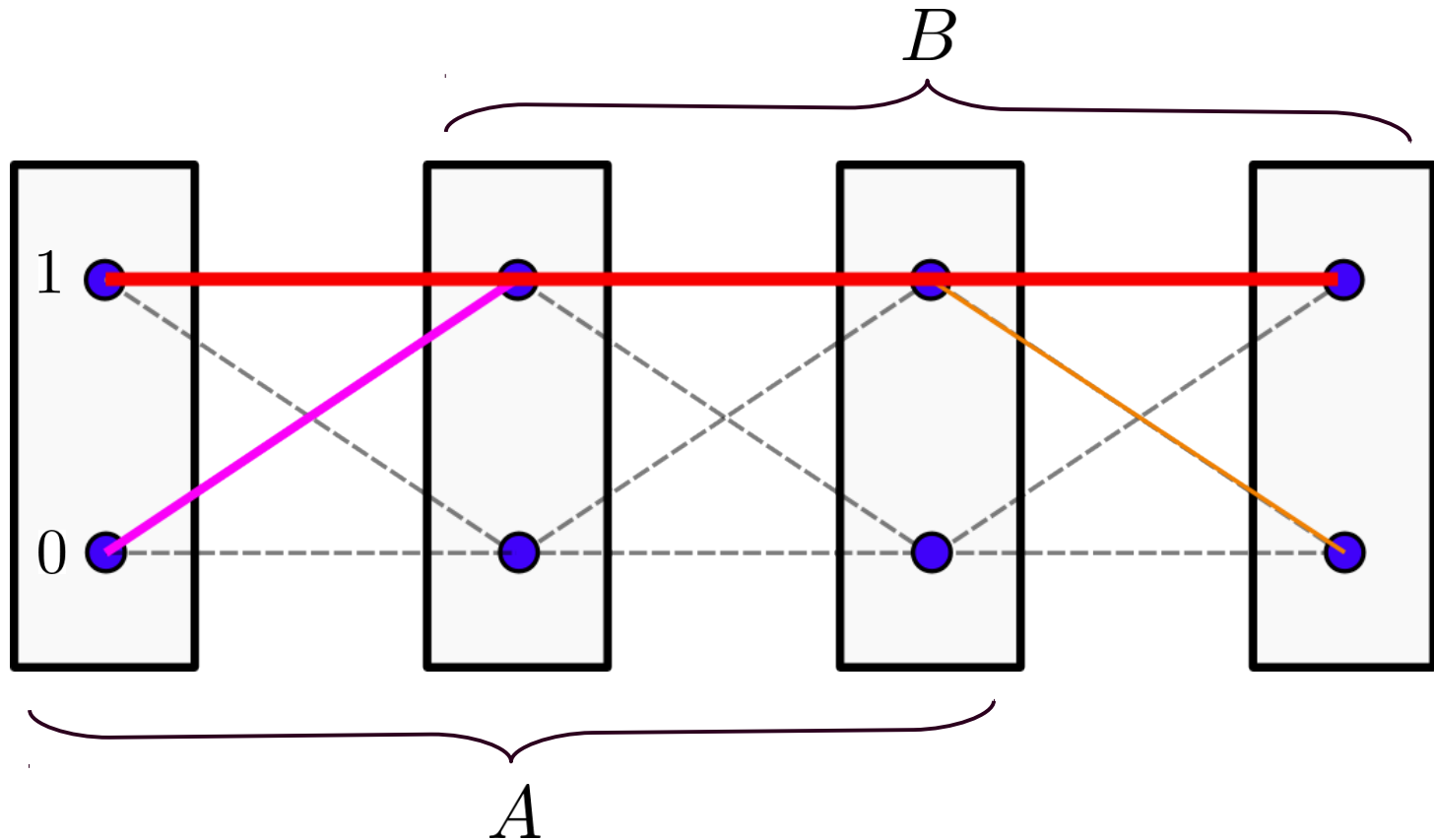
$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

$$B \Leftrightarrow x^B = \{0, \dots, 0, \underbrace{1, \dots, 1}_B\}$$



$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

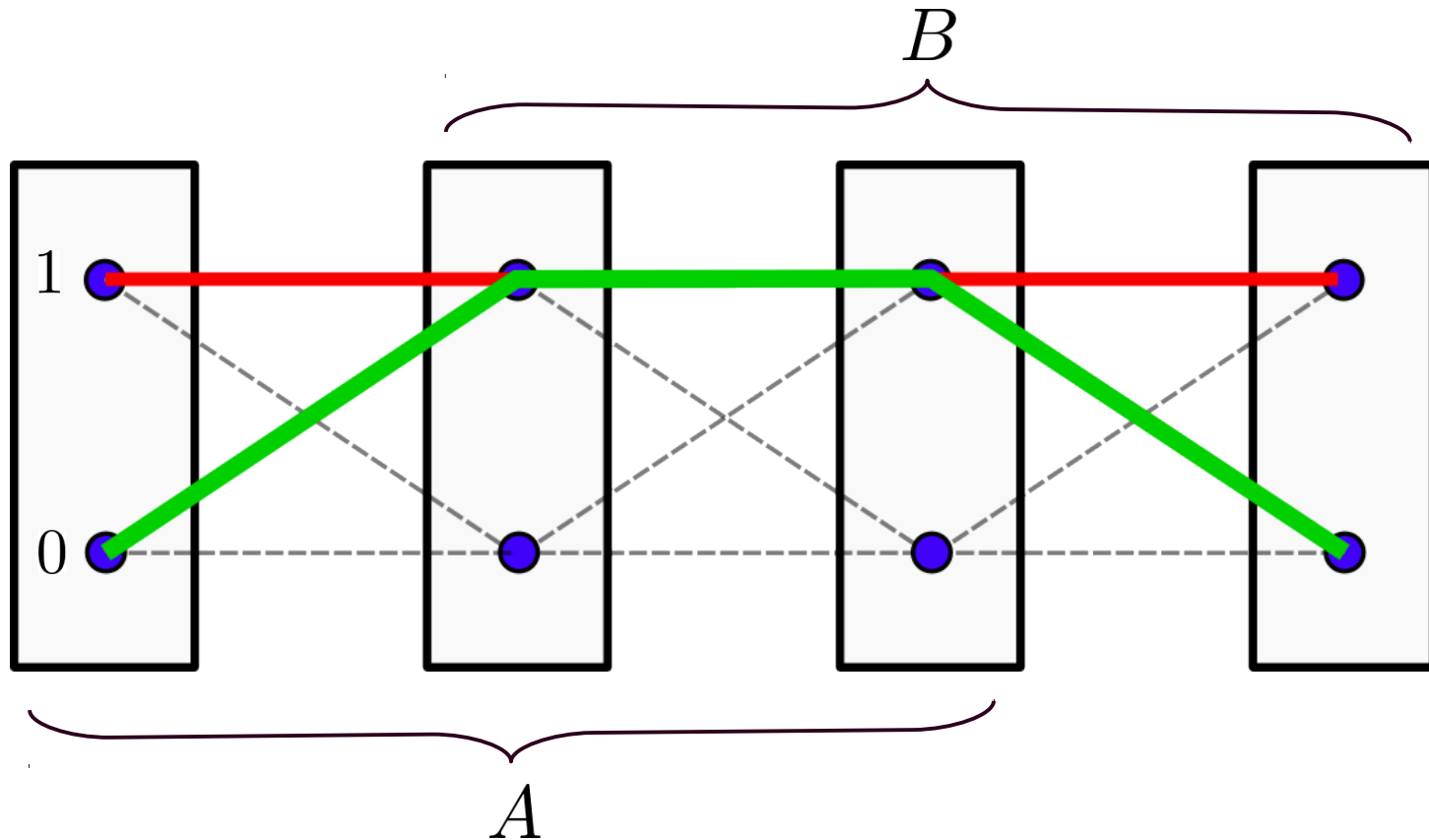
$$\vee \equiv \uparrow$$



$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

$$\vee \equiv \uparrow$$

$$\wedge \equiv \downarrow$$



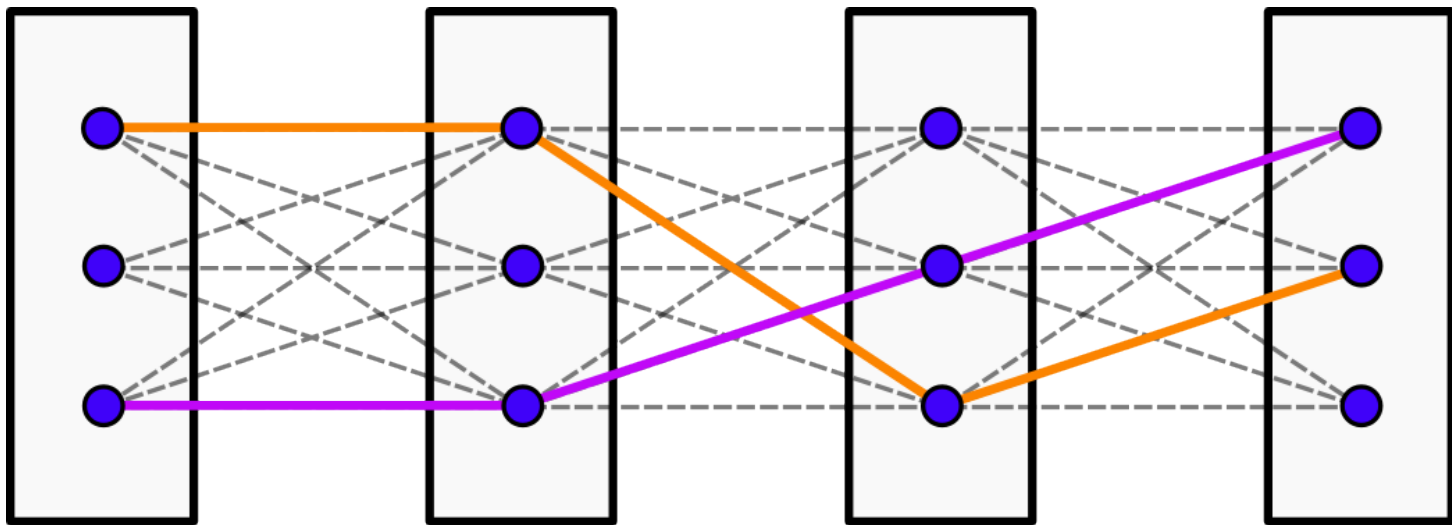
Submodular Energies

We will assume an ordering on the set of labels, $\vee \equiv \max$, $\wedge \equiv \min$

$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

$$\vee \equiv \uparrow$$

$$\wedge \equiv \downarrow$$



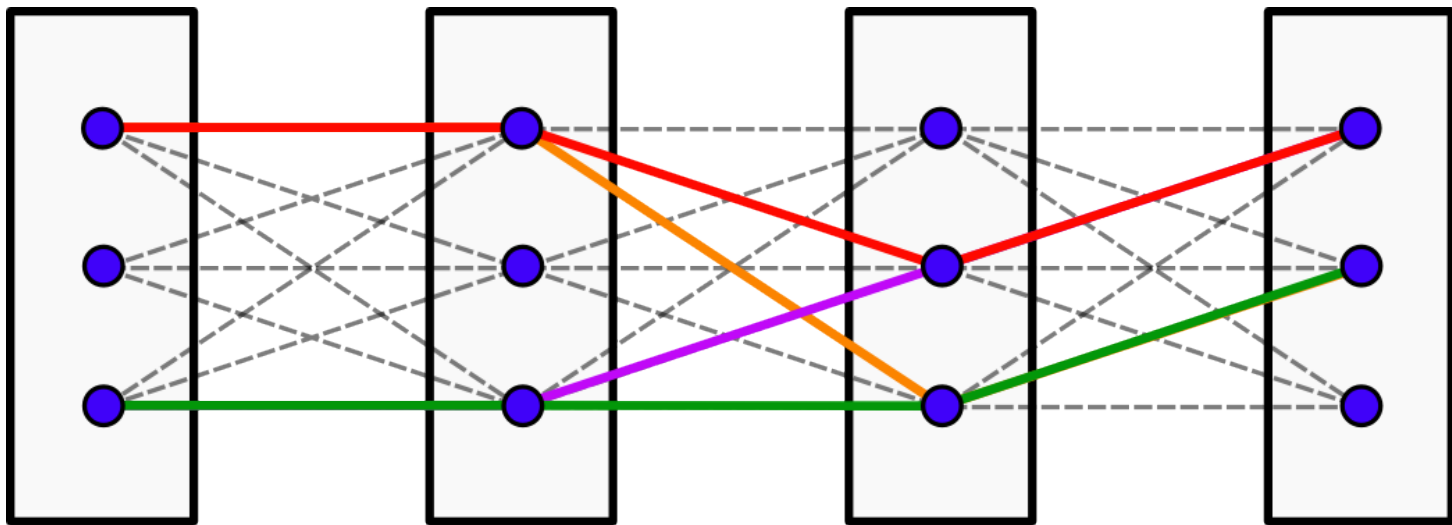
Submodular Energies

We will assume an ordering on the set of labels, $\vee \equiv \max$, $\wedge \equiv \min$

$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

$\vee \equiv \uparrow$

$\wedge \equiv \downarrow$



E – submodular:

$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

E – supermodular:

$$E(x^A) + E(x^B) \leq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

E – submodular:

$$E(x^A) + E(x^B) \geq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

E – supermodular:

$$E(x^A) + E(x^B) \leq E(x^A \vee x^B) + E(x^A \wedge x^B)$$

E – modular:

$$E(x^A) + E(x^B) = E(x^A \vee x^B) + E(x^A \wedge x^B)$$

$$\mathcal{G} = \{\{v\}, \emptyset\}$$

Unary potentials: $\theta_v(s) + \theta_v(t) \stackrel{?}{=} \theta_v(s \vee t) + \theta_v(s \wedge t), \quad s, t \in \mathcal{X}_v$

Unary potentials are:

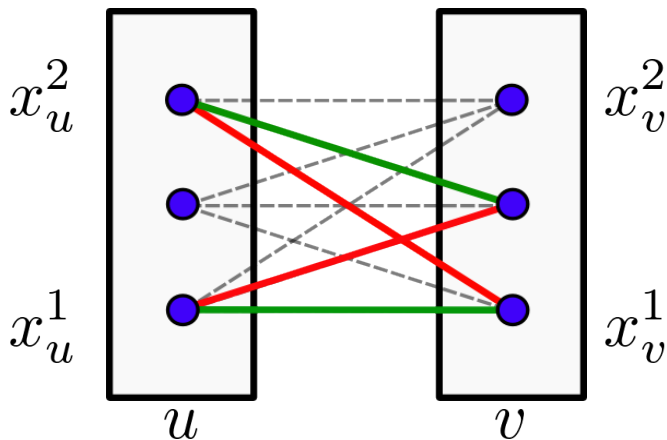
- 1) submodular
- 2) supermodular
- 3) modular
- 4) neither of 1)-3) holds in general
- 5) 1)-3) are all correct
- 6) no idea

Unary potentials are always modular:

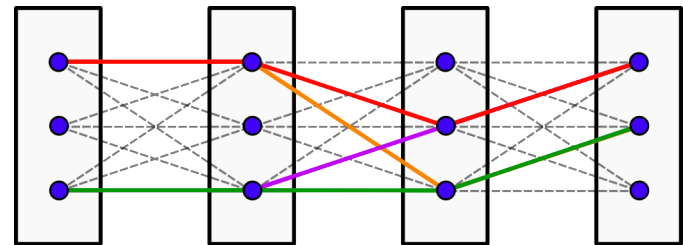
$$\theta_v(s) + \theta_v(t) = \theta_v(s \vee t) + \theta_v(s \wedge t), \quad s, t \in \mathcal{X}_v$$

Pairwise potentials:

$$\theta_{uv}(x_u^1, x_v^1) + \theta_{uv}(x_u^2, x_v^2) \leq \theta_{uv}(x_u^2, x_v^1) + \theta_{uv}(x_u^1, x_v^2)$$

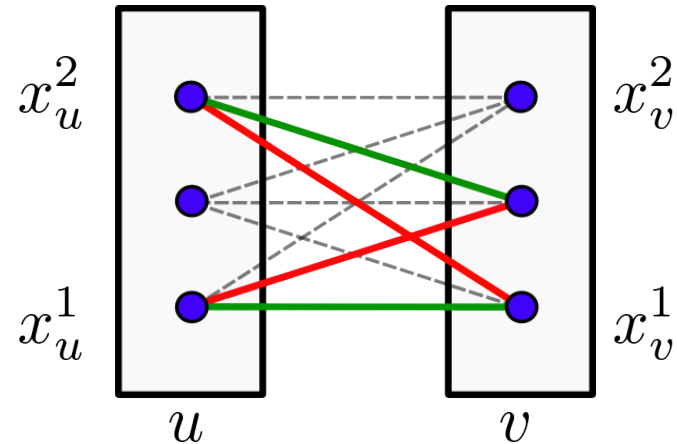


Compare to:



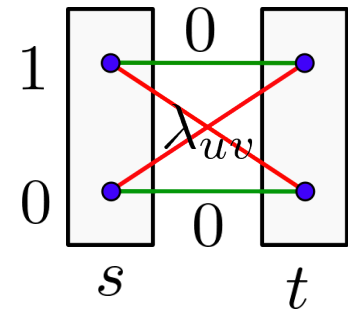
Submodular Pairwise Potentials: Examples

Pairwise potentials:



Ising model: 2 labels, $\theta_{uv}(s, t) = \lambda_{uv} \mathbb{I}[s \neq t]$

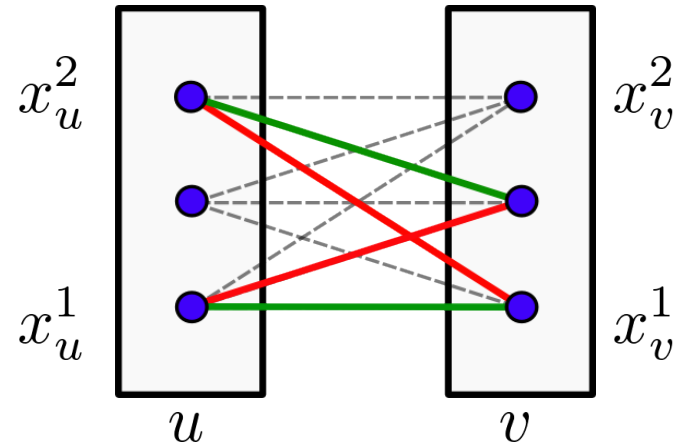
- 1) submodular for $\lambda_{uv} \geq 0$
- 2) submodular for $\lambda_{uv} \leq 0$
- 3) modular for any λ_{uv}
- 4) neither is correct
- 5) no idea



Application: foreground/background segmentation

Submodular Pairwise Potentials: Examples

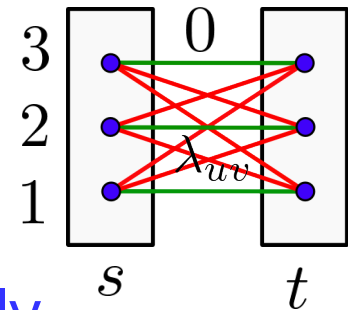
Pairwise potentials:



Potts model:

$$\theta_{uv}(s, t) = \lambda_{uv} \mathbb{I}[s \neq t]$$

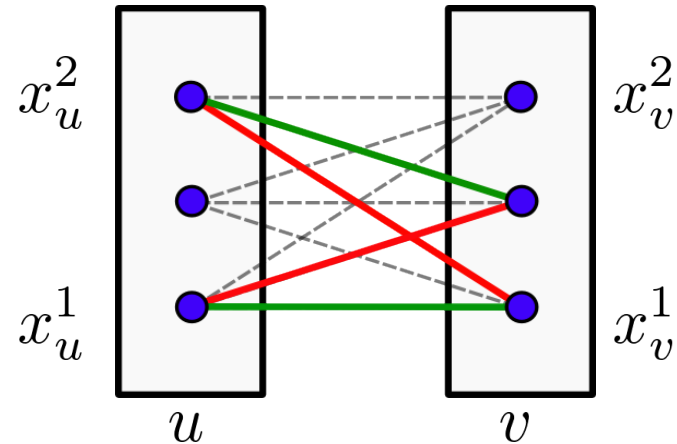
- 1) submodular for $\lambda_{uv} \geq 0$
- 2) submodular for $\lambda_{uv} \leq 0$
- 3) submodular for $\lambda_{uv} \geq 0$ and 2 labels only
- 4) modular for any λ_{uv}
- 5) neither is correct
- 6) no idea



Application: semantic segmentation

Submodular Pairwise Potentials: Examples

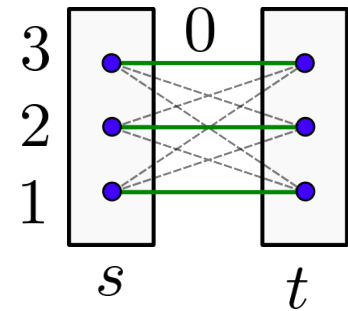
Pairwise potentials:



Convex potentials: $\mathcal{X}_v = \{1, \dots, n\}$

$$\theta_{uv}(s, t) = |s - t|$$

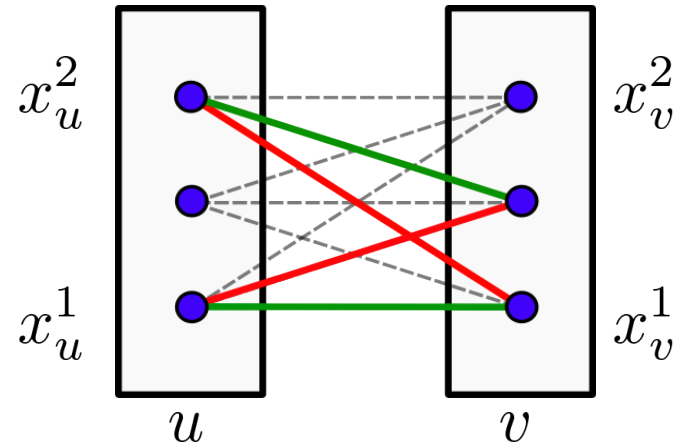
$$\theta_{uv}(s, t) = (s - t)^2$$



Application: stereo, optical flow

Exercise: Prove that convex pairwise potentials are submodular

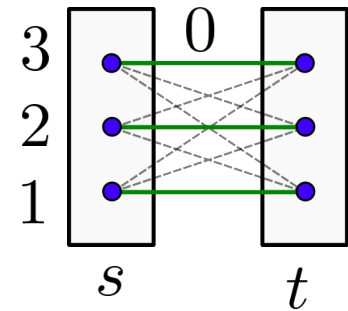
Pairwise potentials:



Truncated-convex potentials: $\mathcal{X}_v = \{1, \dots, n\}$

$$\theta_{uv}(s, t) = \min\{a, |s - t|\}$$

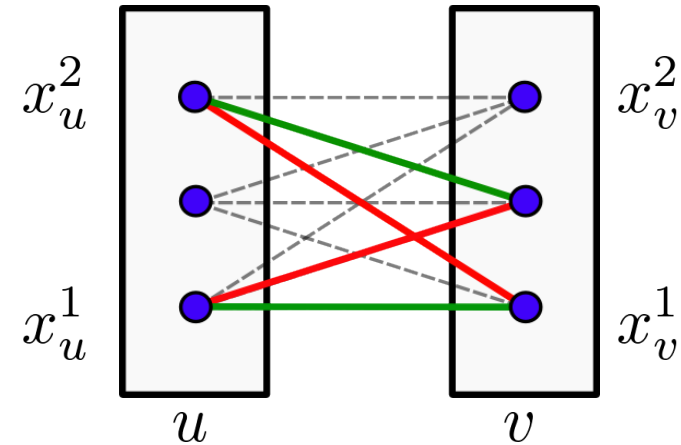
$$\theta_{uv}(s, t) = \min\{a, (s - t)^2\}$$



Application: robust stereo, optical flow

Submodular Pairwise Potentials: Examples

Pairwise potentials:

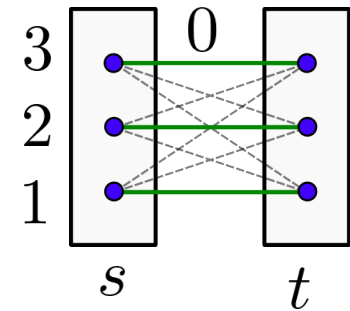


Truncated-convex potentials: $\mathcal{X}_v = \{1, \dots, n\}$

$$\theta_{uv}(s, t) = \min\{a, |s - t|\}$$

$$a > 0$$

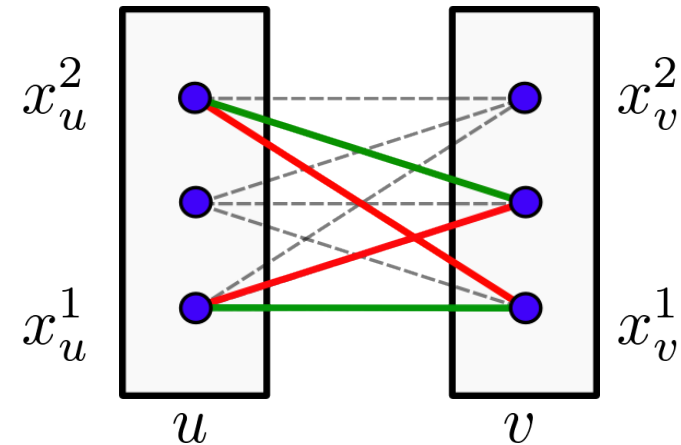
$$\theta_{uv}(s, t) = \min\{a, (s - t)^2\}$$



- | | |
|-----------------|--------------------|
| 1) submodular | 4) none is correct |
| 2) supermodular | 5) no idea |
| 3) modular | |

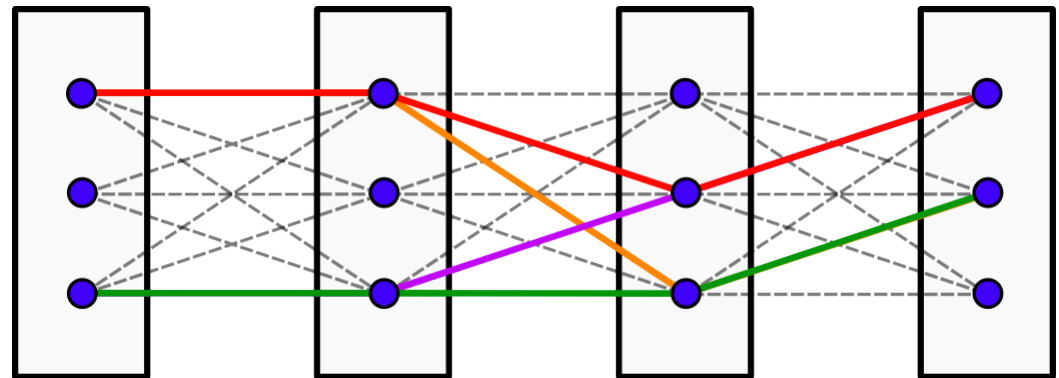
Submodular Potentials vs. Submodular Energy

Submodular pairwise potentials:



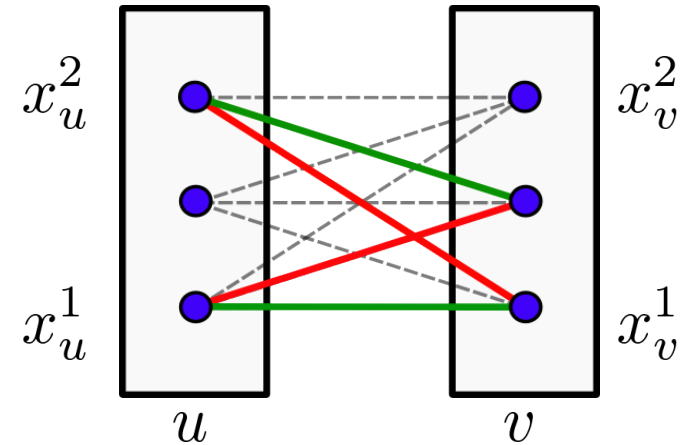
Exercise: *Sum of submodular functions is submodular*

Submodular energy:

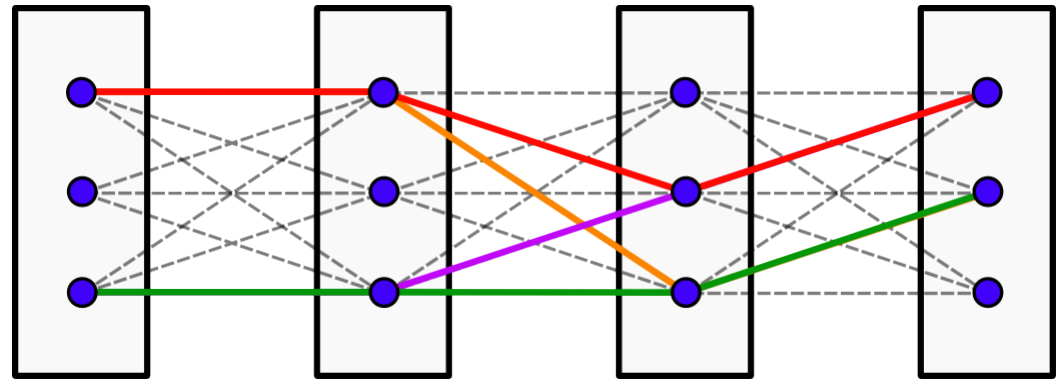


Submodular Potentials vs. Submodular Energy

Submodular pairwise potentials:

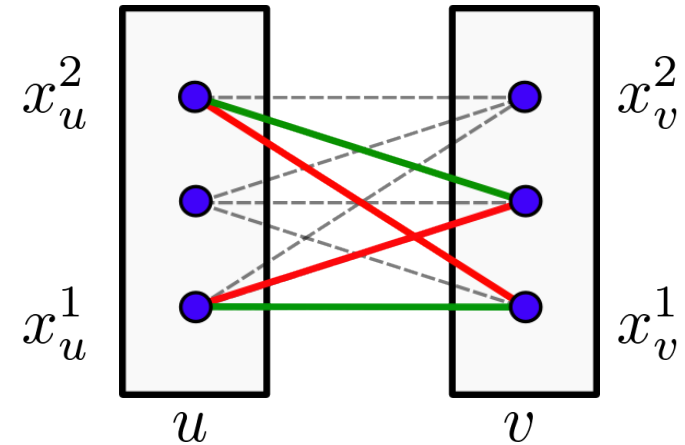


Submodular energy:



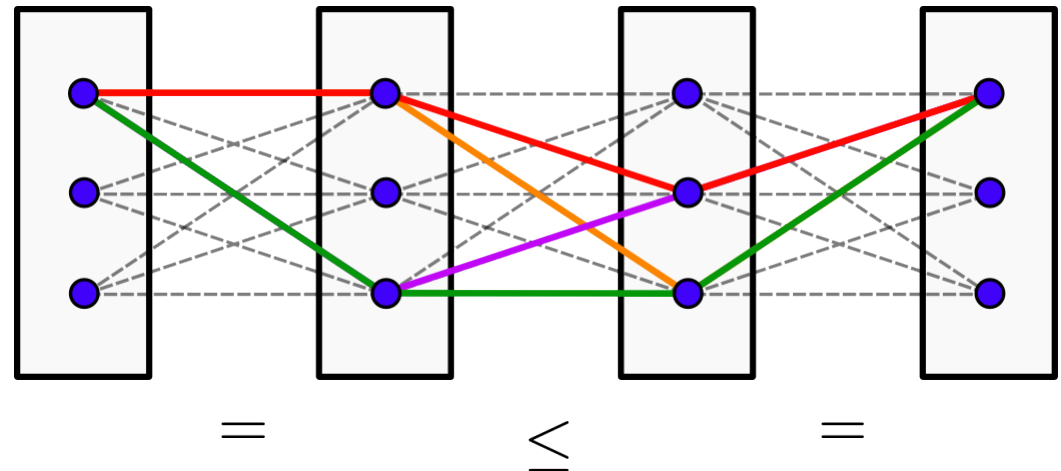
Submodular Potentials vs. Submodular Energy

Submodular pairwise potentials:



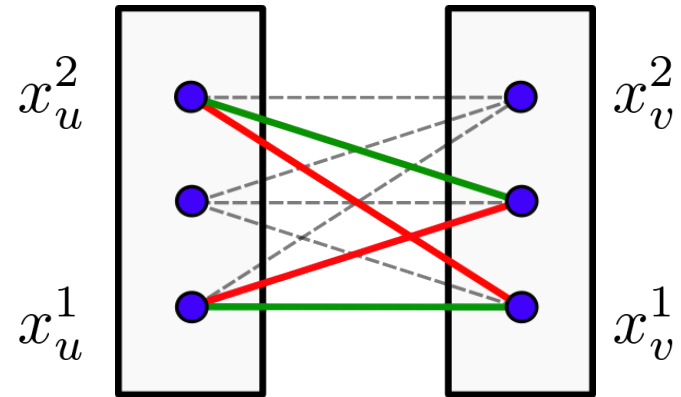
Note: these arguments do not hold in higher order case

Submodular energy:

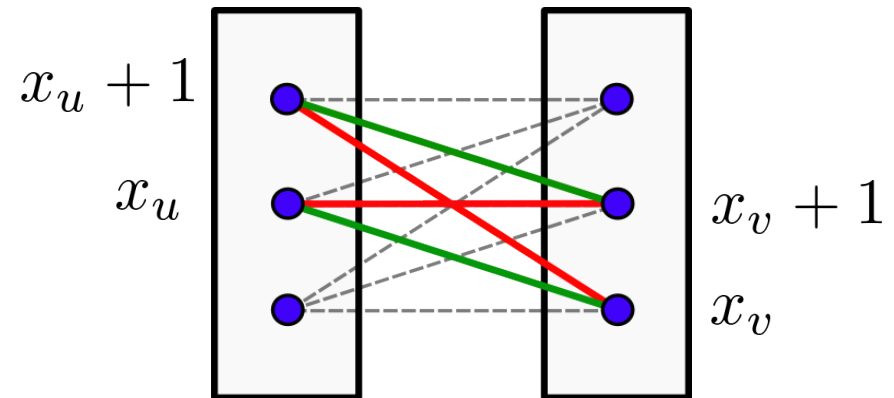


Submodular Pairwise Potentials: Linear-Time Proof

$$\begin{aligned} \theta_{uv}(x_u^1, x_v^1) + \theta_{uv}(x_u^2, x_v^2) \\ \leq \theta_{uv}(x_u^2, x_v^1) + \theta_{uv}(x_u^1, x_v^2) \\ O(L^4) \end{aligned}$$



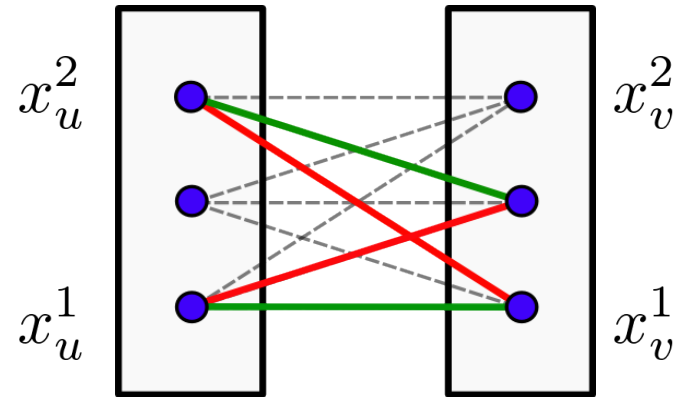
$$O(L^2)$$



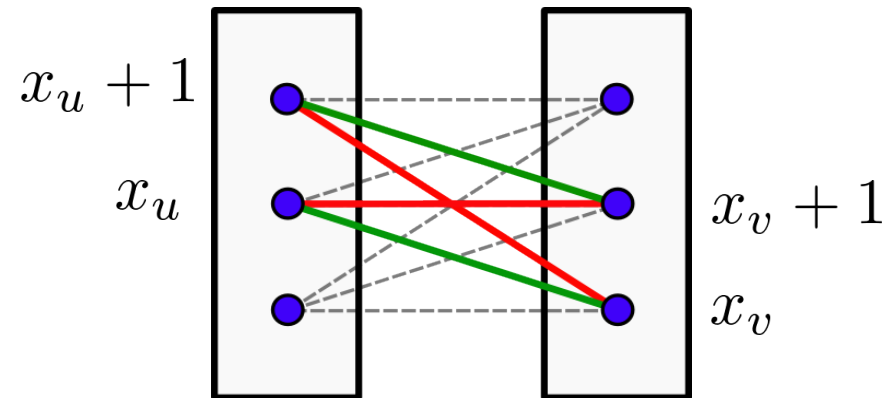
$$\begin{aligned} \theta_{uv}(x_u, x_v) + \theta_{uv}(x_u + 1, x_v + 1) \\ \leq \theta_{uv}(x_u + 1, x_v) + \theta_{uv}(x_u, x_v + 1) \end{aligned}$$

Submodular Pairwise Potentials: Linear-Time Proof

$$\begin{aligned} \theta_{uv}(x_u^1, x_v^1) + \theta_{uv}(x_u^2, x_v^2) \\ \leq \theta_{uv}(x_u^2, x_v^1) + \theta_{uv}(x_u^1, x_v^2) \\ O(L^4) \end{aligned}$$



$$O(L^2)$$

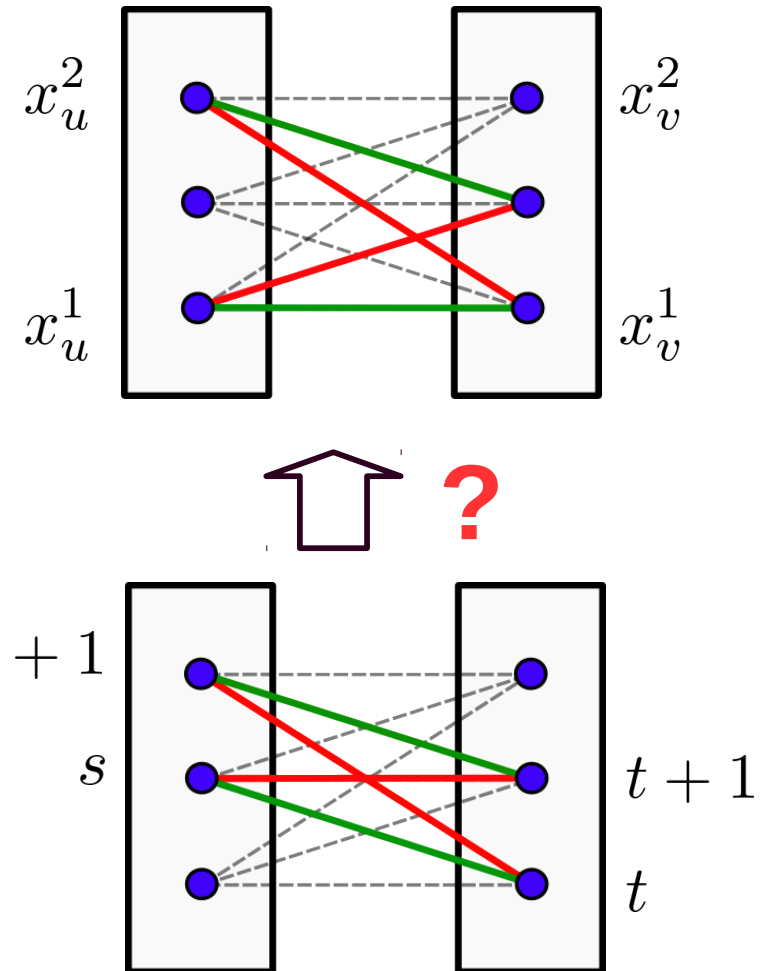


$$\begin{aligned} \theta_{uv}(x_u, x_v) + \theta_{uv}(x_u + 1, x_v + 1) \\ \leq \theta_{uv}(x_u + 1, x_v) + \theta_{uv}(x_u, x_v + 1) \end{aligned}$$

Submodular Pairwise Potentials: Linear-Time Proof

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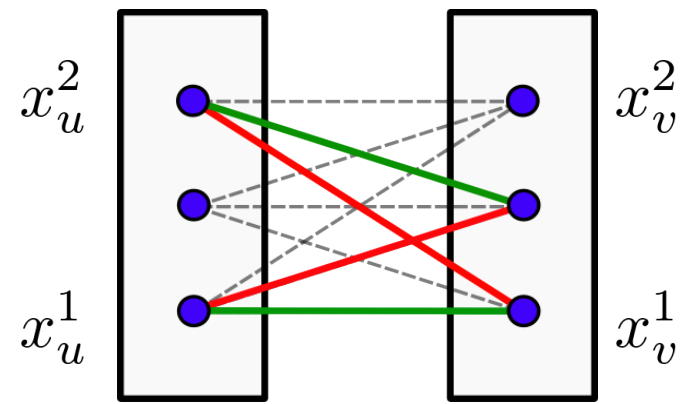
$$f(a) - f(b) = \sum_{i=b}^{a-1} f(i+1) - f(i)$$



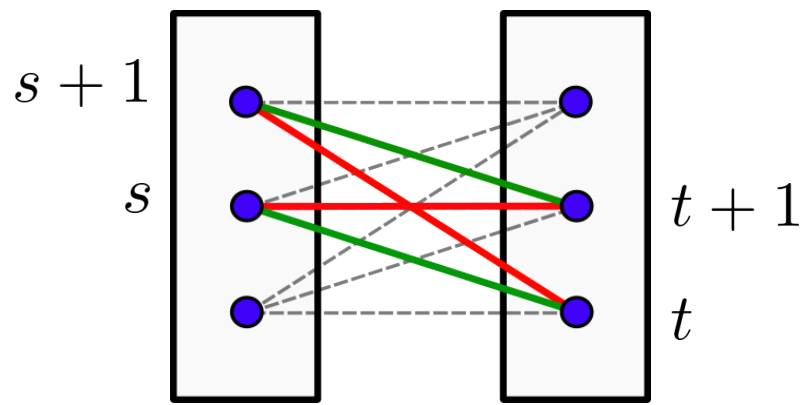
$$\begin{aligned} \theta_{uv}(s, t) + \theta_{uv}(s+1, t+1) \\ \leq \theta_{uv}(s+1, t) + \theta_{uv}(s, t+1) \end{aligned}$$

Submodular Pairwise Potentials: Linear-Time Proof

$$\begin{aligned} \theta_{uv}(x_u^1, x_v^1) + \theta_{uv}(x_u^2, x_v^2) \\ \leq \theta_{uv}(x_u^2, x_v^1) + \theta_{uv}(x_u^1, x_v^2) \end{aligned}$$

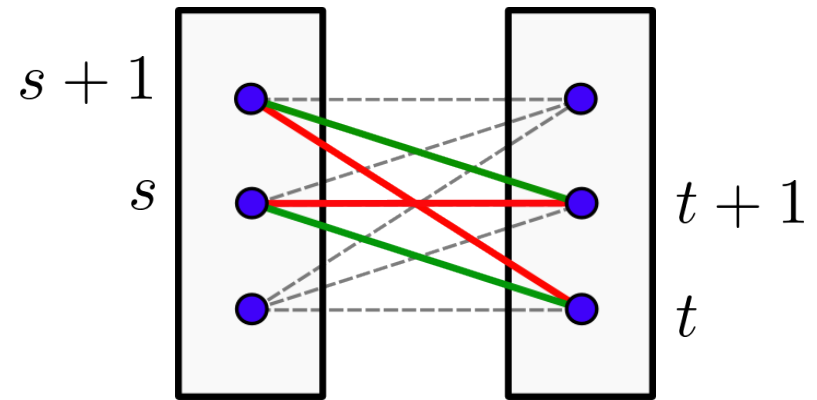
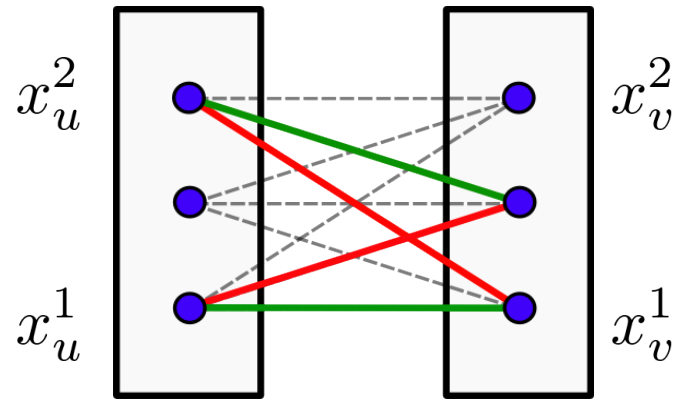


$$\begin{aligned} f(a) - f(b) &= \sum_{i=b}^{a-1} f(i+1) - f(i) \\ f(a) - f(b) &= \int_b^a df = \int_b^a f'(t) dt \end{aligned}$$



$$\begin{aligned} \theta_{uv}(s, t) + \theta_{uv}(s+1, t+1) \\ \leq \theta_{uv}(s+1, t) + \theta_{uv}(s, t+1) \end{aligned}$$

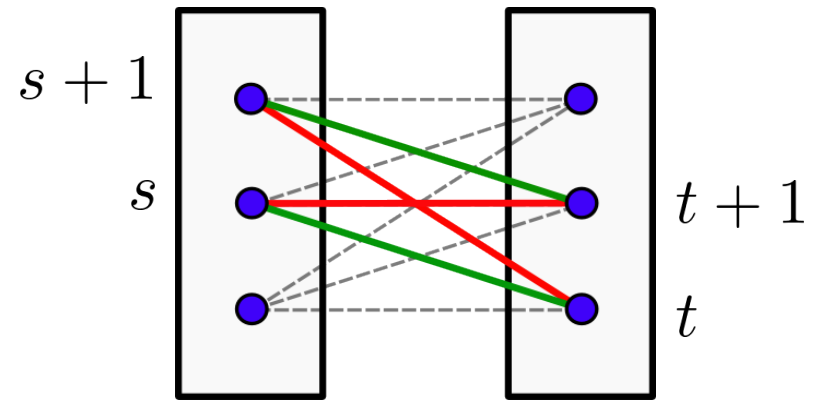
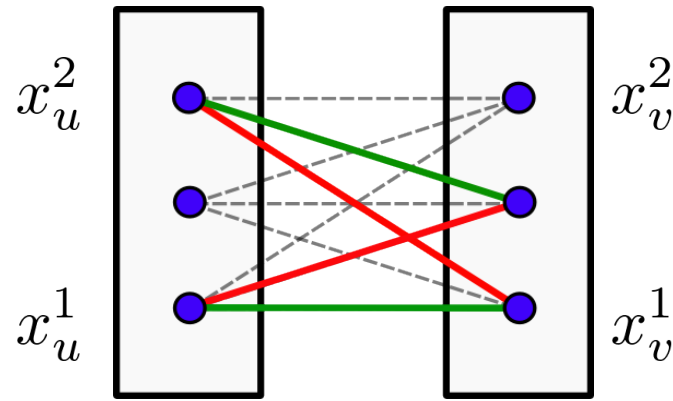
Submodular Pairwise Potentials: Derivative



$$f(a, b) + f(a', b') - f(a', b) - f(a, b') = \int_a^{a'} \int_b^{b'} \frac{\partial^2 f(s, t)}{\partial s \partial t} ds dt$$

$$\frac{\partial^2 \theta_{uv}(s, t)}{\partial s \partial t} = \theta_{uv}(s, t) + \theta_{uv}(s + 1, t + 1) - \theta_{uv}(s, t + 1) - \theta_{uv}(s + 1, t)$$

Submodular Pairwise Potentials: Derivative

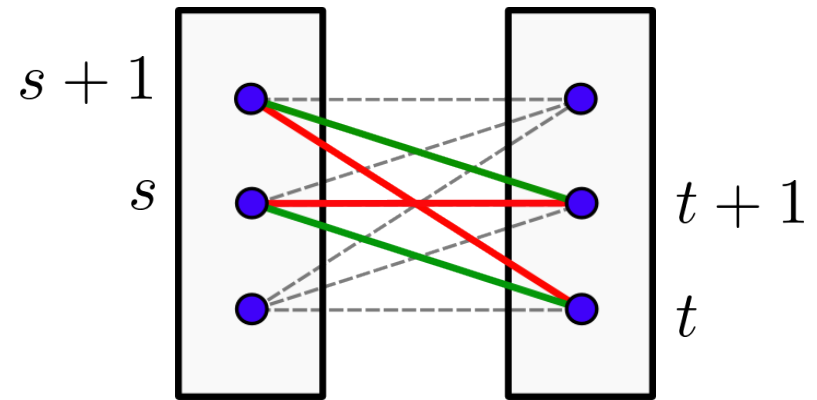
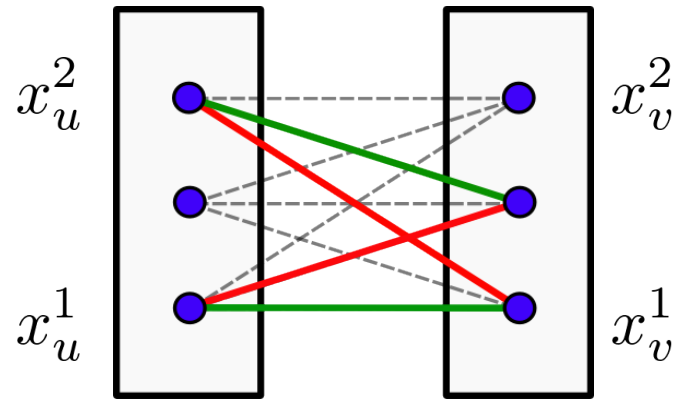


$$f(a, b) + f(a', b') - f(a', b) - f(a, b') = \int_a^{a'} \int_b^{b'} \frac{\partial^2 f(s, t)}{\partial s \partial t} ds dt$$

$$\theta(x^2, z^2) + \theta(x^1, z^1) - \theta(x^1, z^2) - \theta(x^2, z^1) = \sum_{s=x^1}^{x^2-1} \sum_{t=z^1}^{z^2-1} \frac{\partial^2 \theta(s, t)}{\partial s \partial t}$$

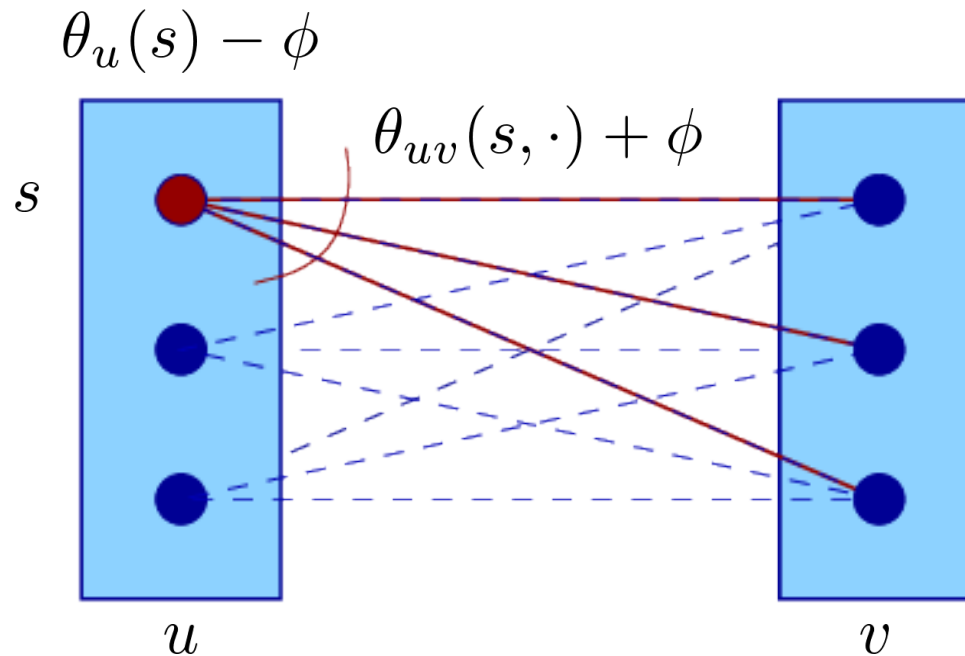
$$\frac{\partial^2 \theta_{uv}(s, t)}{\partial s \partial t} = \theta_{uv}(s, t) + \theta_{uv}(s+1, t+1) - \theta_{uv}(s, t+1) - \theta_{uv}(s+1, t)$$

Submodular Pairwise Potentials: Derivative



(Another) Definition: θ_{uv} is submodular iff $\frac{\partial^2 \theta_{uv}(s, t)}{\partial s \partial t} \leq 0$.

Properties of Submodular Energies: Reparametrization

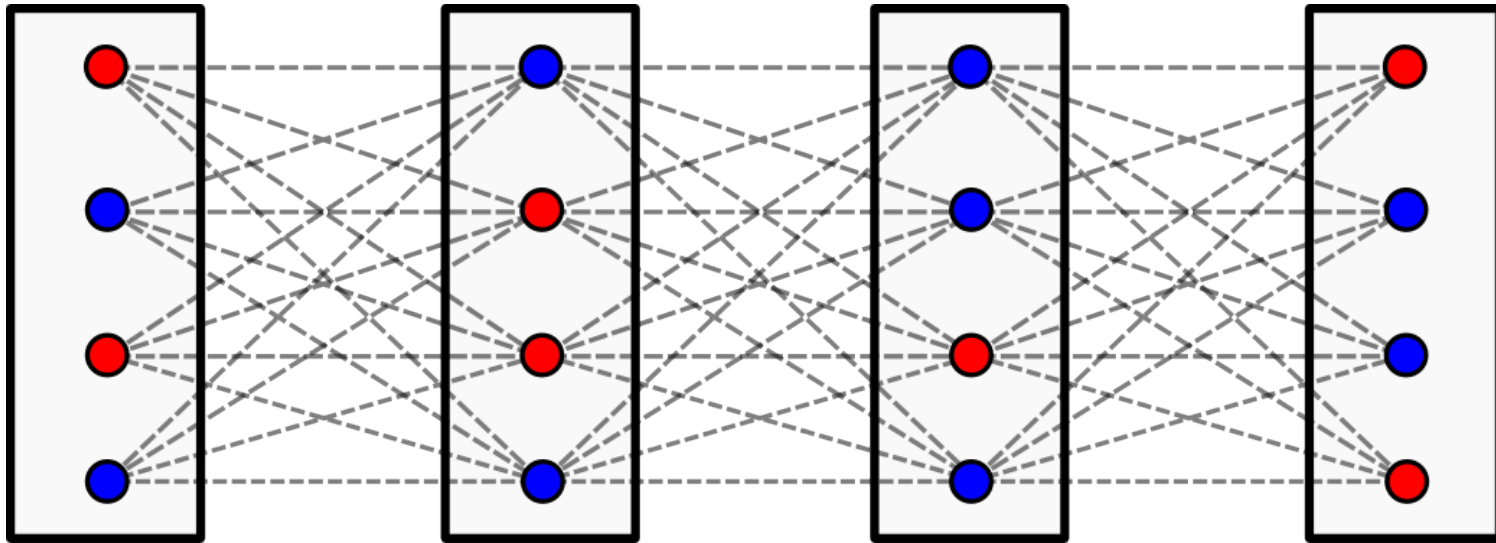


Let θ be submodular. Then $\theta^\phi \dots$

- 1) Is submodular as well
- 2) is not submodular in general
- 3) no idea

Submodular Energies: Properties

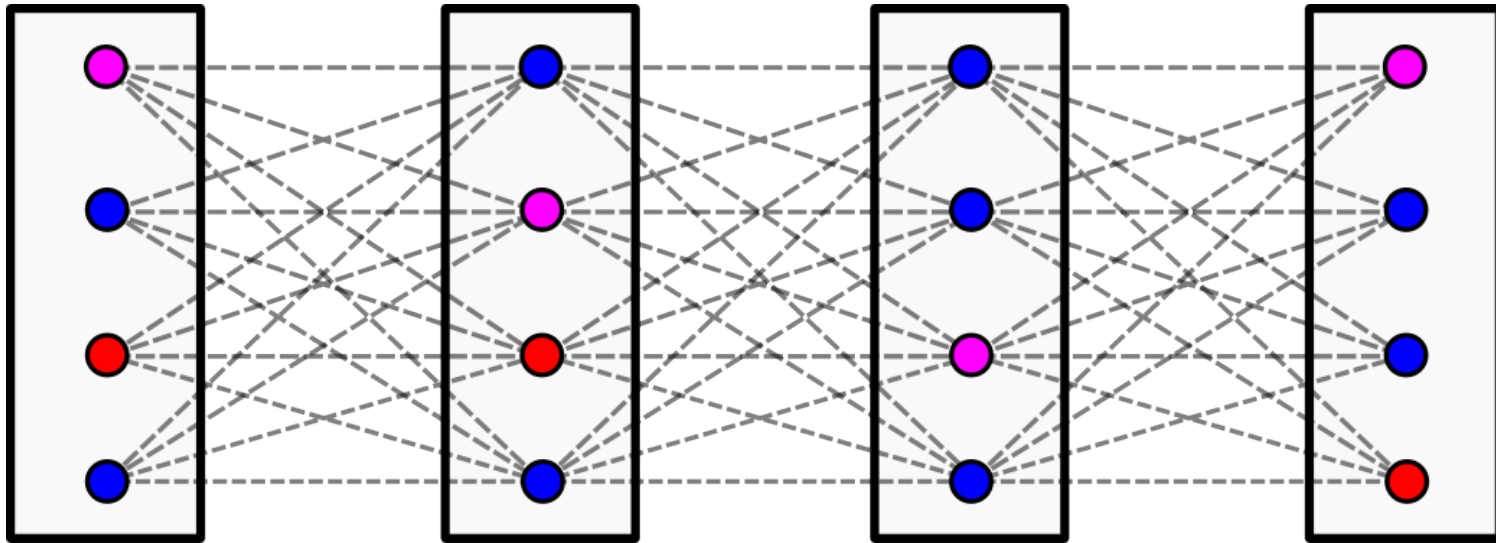
Theorem. Let θ be submodular and $\text{kern}(\theta^\phi) \neq \emptyset$.
Then there exists an integer labeling x such that $E(\theta, x) = D(\phi)$
(i.e. ϕ is optimal and LP relaxation is tight)



Submodular Energies: Properties

Theorem. Let θ be submodular and $\text{kern}(\theta^\phi) \neq \emptyset$.
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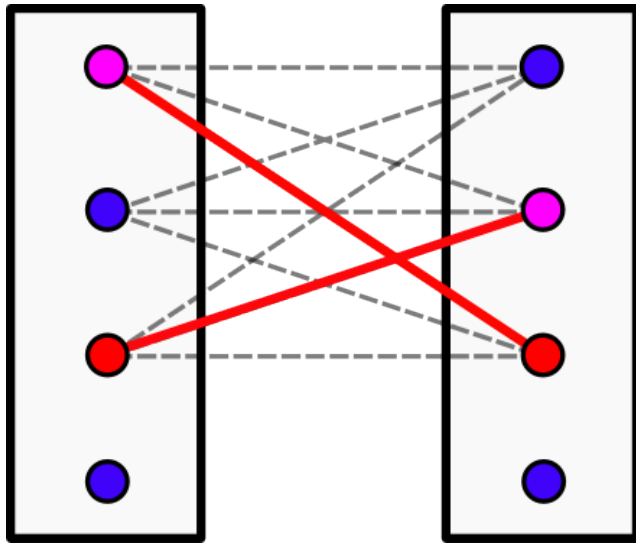
Select the highest label in each node...



Submodular Energies: Properties

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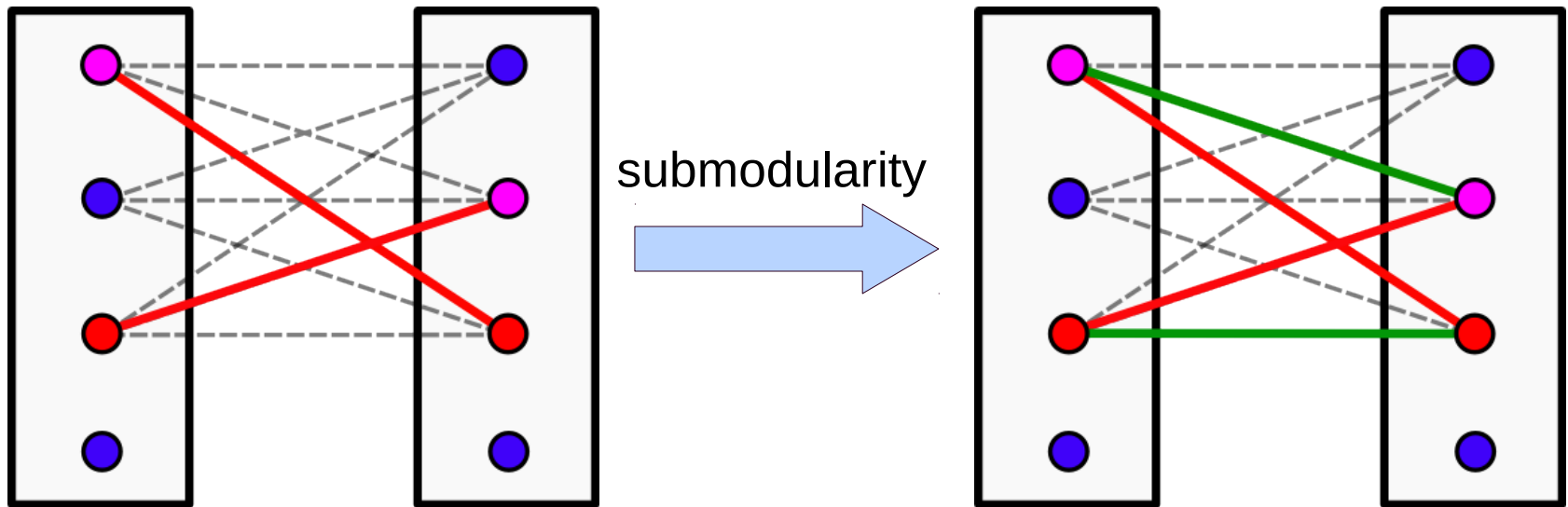
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Submodular Energies: Properties

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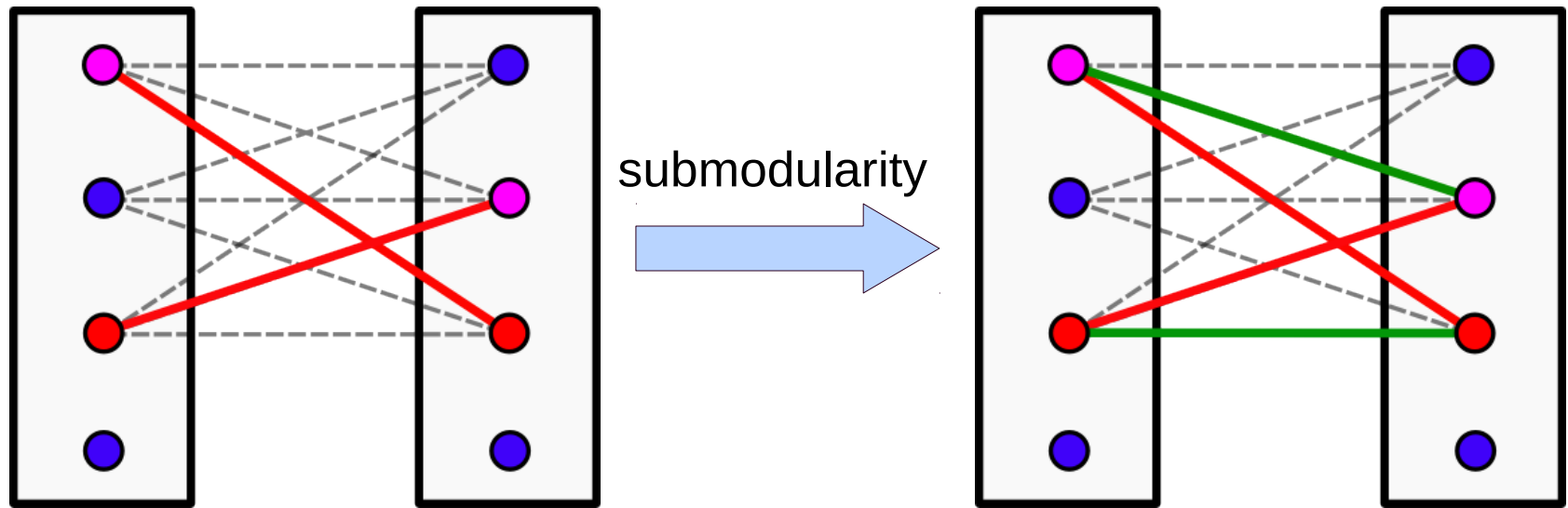
Select the highest label in each node...



Submodular Energies: Properties

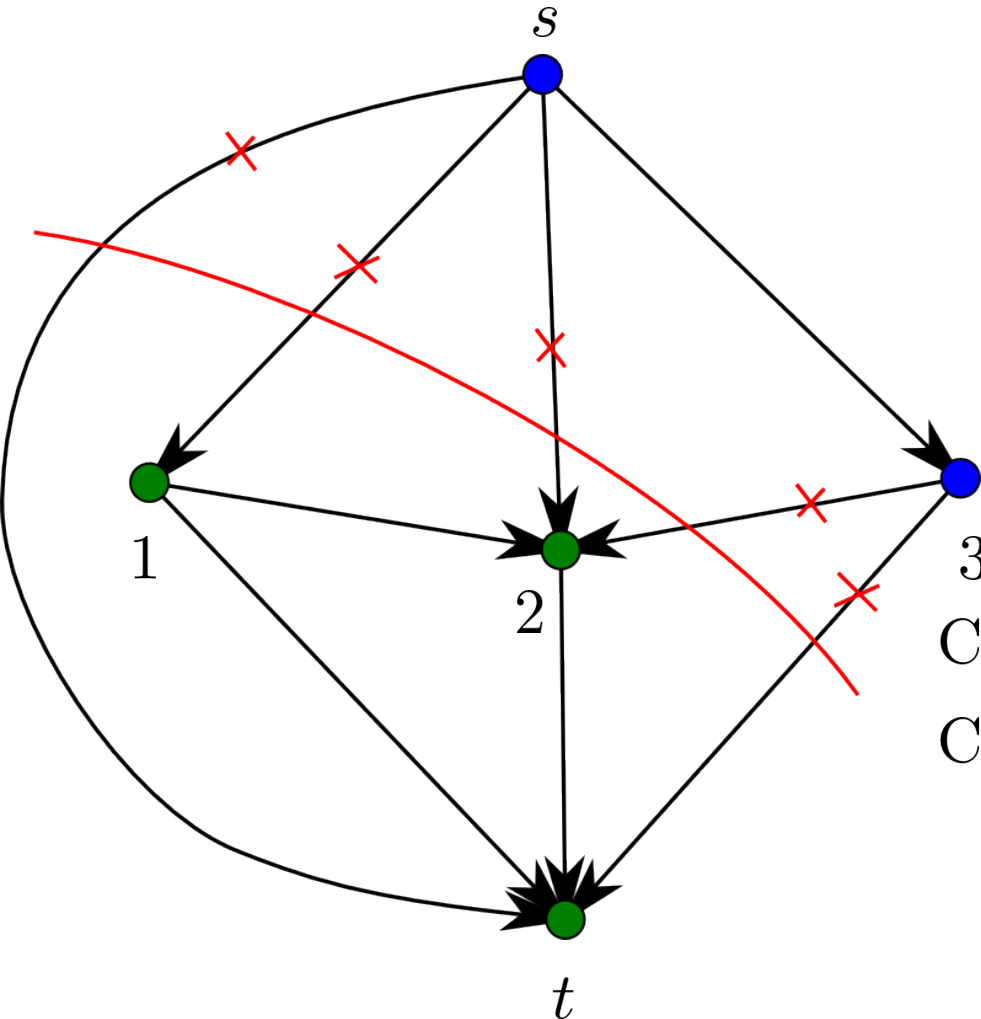
Theorem. Let θ be submodular and $\text{kern}(\theta^\phi) \neq \emptyset$.
Then there exists an integer labeling x such that $E(\theta, x) = D(\phi)$
(i.e. ϕ is optimal and LP relaxation is tight)

Select the highest label in each node...



Corollary: Submodular problems are exactly solvable with Min-Sum Diffusion and TRW-S.

st-Min-Cut Problem



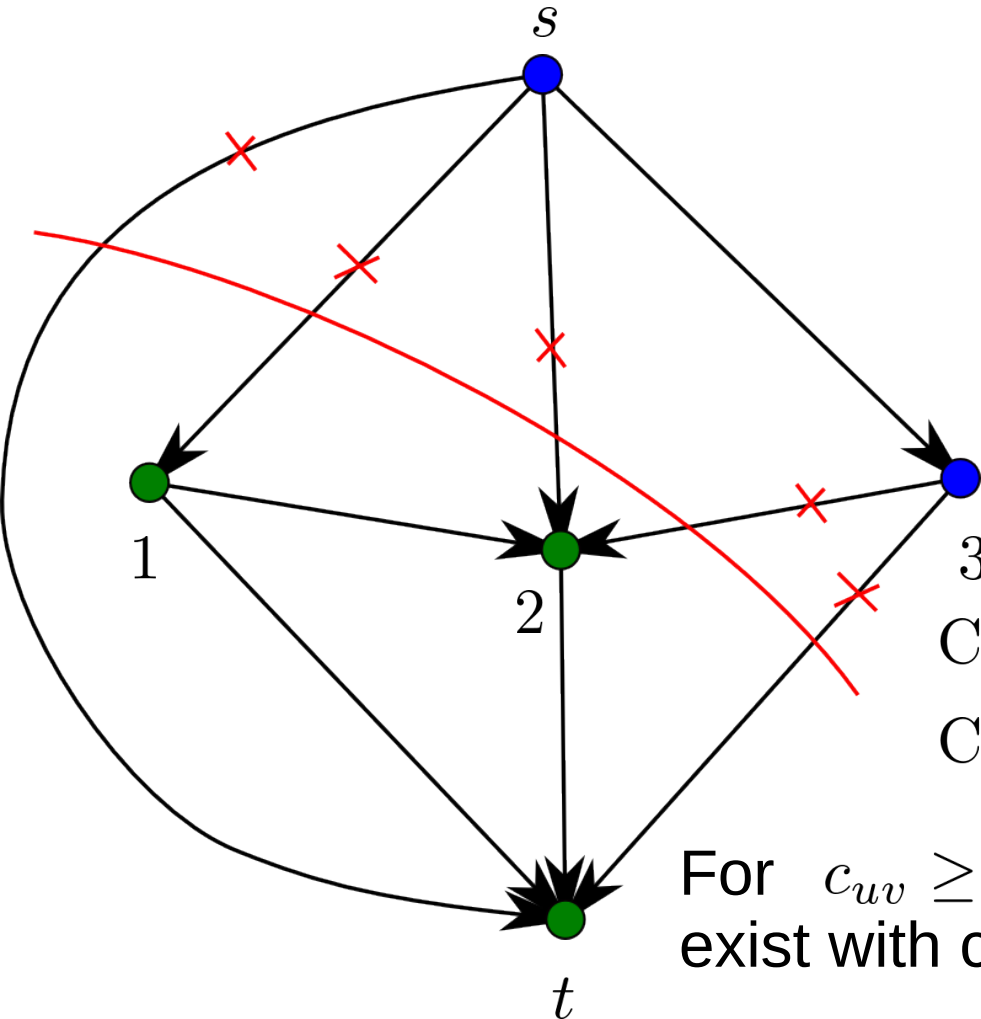
st-Min-cut problem:

$$\min_{S,T} \sum_{\substack{u \in S, v \in T \\ uv \in \mathcal{E}'}} c_{uv}$$

Cut: $(s, t), (s, 1), (s, 2), (3, 2), (3, t)$

Cut cost: $c_{st}, c_{s1}, c_{s2}, c_{32}, c_{3t}$

st-Min-Cut Problem



st-Min-cut problem:

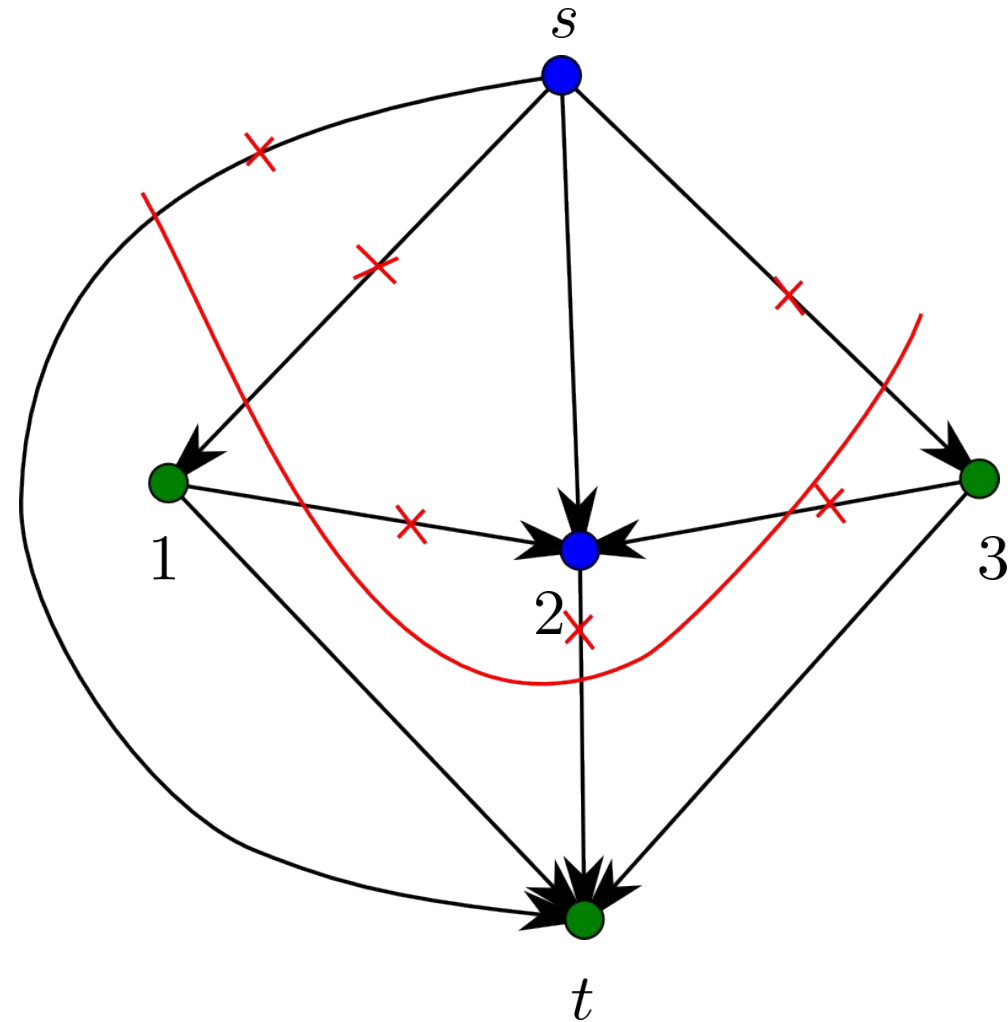
$$\min_{S,T} \sum_{\substack{u \in S, v \in T \\ uv \in \mathcal{E}'}} c_{uv}$$

Cut: $(s, t), (s, 1), (s, 2), (3, 2), (3, t)$

Cut cost: $c_{st}, c_{s1}, c_{s2}, c_{32}, c_{3t}$

For $c_{uv} \geq 0$ efficient finite step algorithms exist with complexity $\approx O(V^3)$

Otherwise – NP-hard



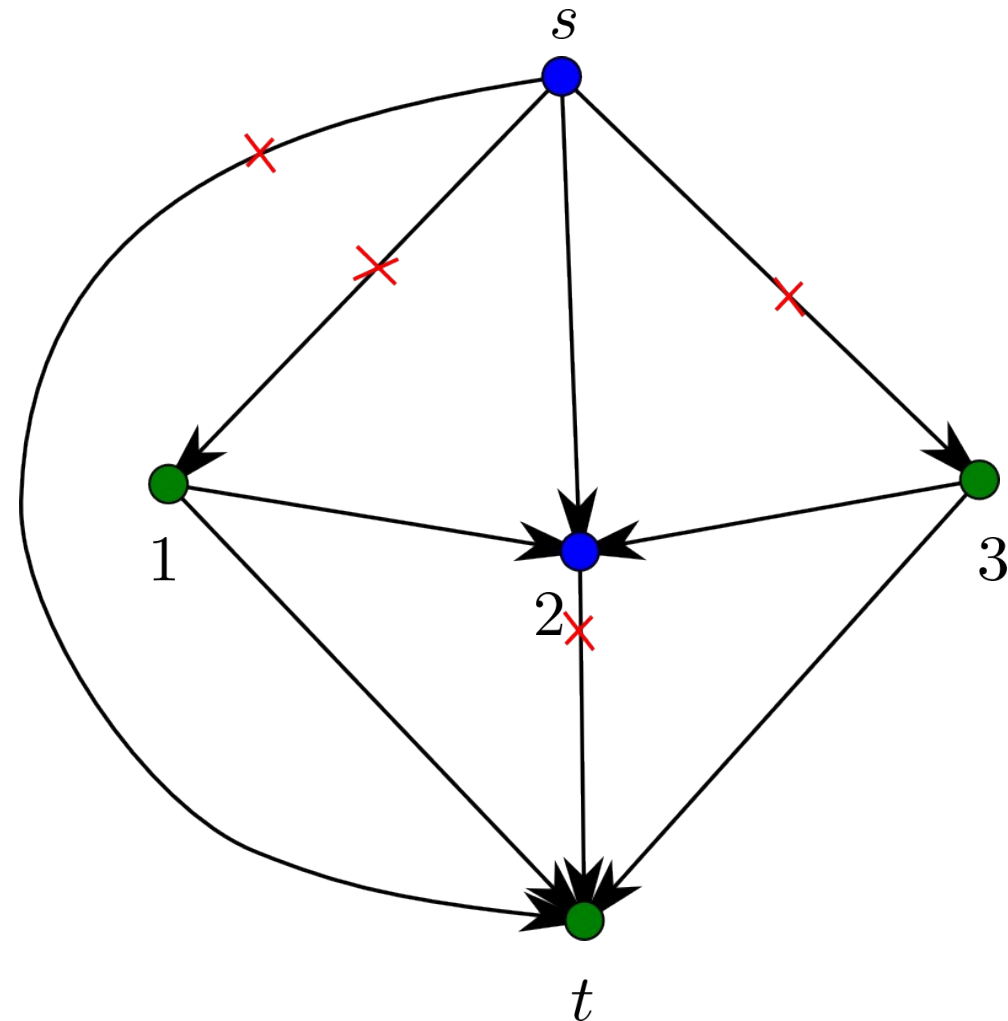
st-Min-cut problem:

$$\min_{S,T} \sum_{\substack{u \in S, v \in T \\ uv \in \mathcal{E}'}} c_{uv}$$

Is it a cut?

- 1) yes
- 2) no, because (1,2) does not belong to it
- 3) no, because (3,2) does not belong to it
- 4) no, because (1,2) and (3,2) does not belong to it
- 5) no, because (2,t) does not belong to it
- 6) none is correct

Cut as Oriented Quadratic Pseudo-Boolean Function



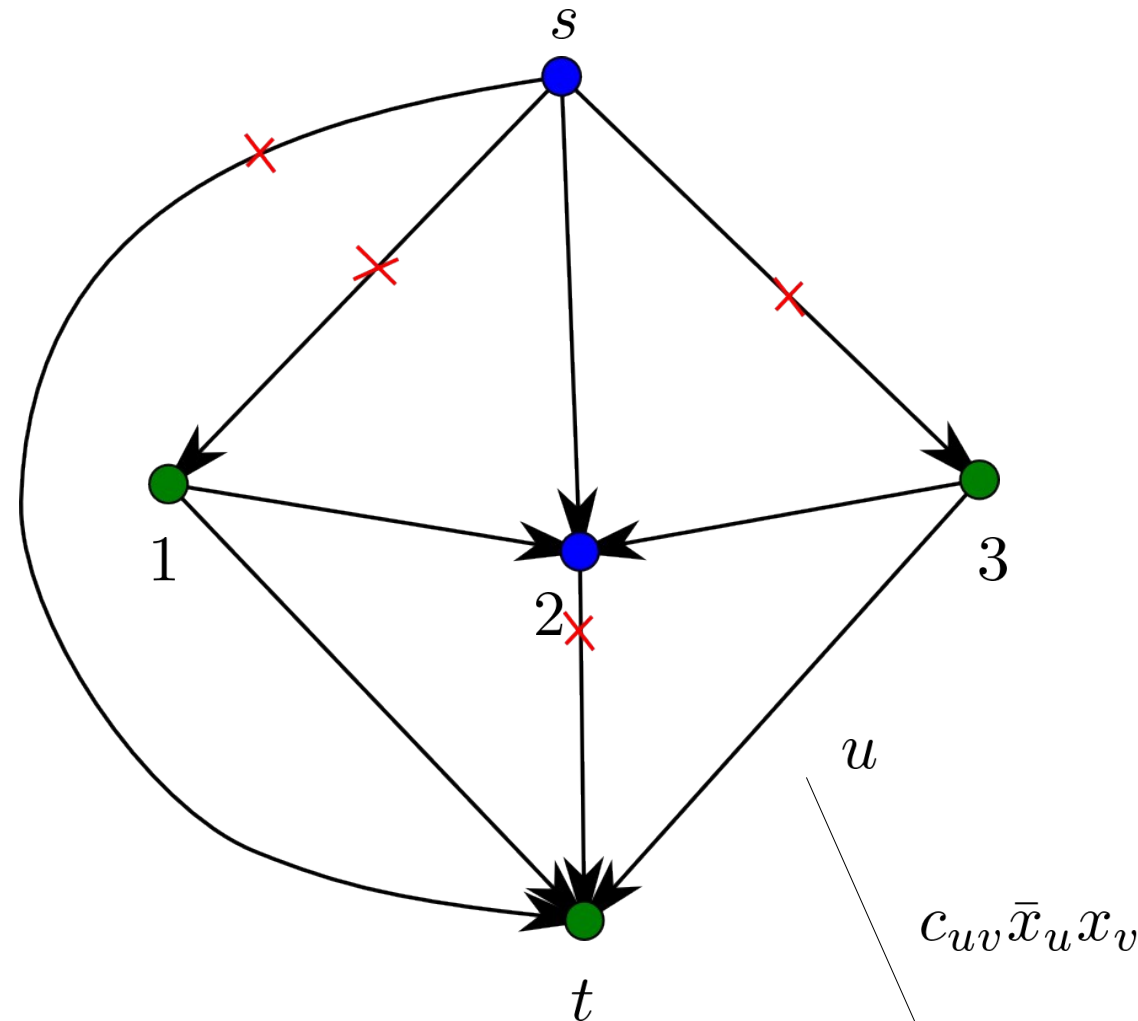
st -Min-cut problem:

$$\min_{S,T} \sum_{\substack{u \in S, v \in T \\ uv \in \mathcal{E}'}} c_{uv}$$

$$(x_i = 0) \Rightarrow (i \in S)$$

$$(x_i = 1) \Rightarrow (i \in T)$$

Cut as Oriented Quadratic Pseudo-Boolean Function



st-Min-cut problem:

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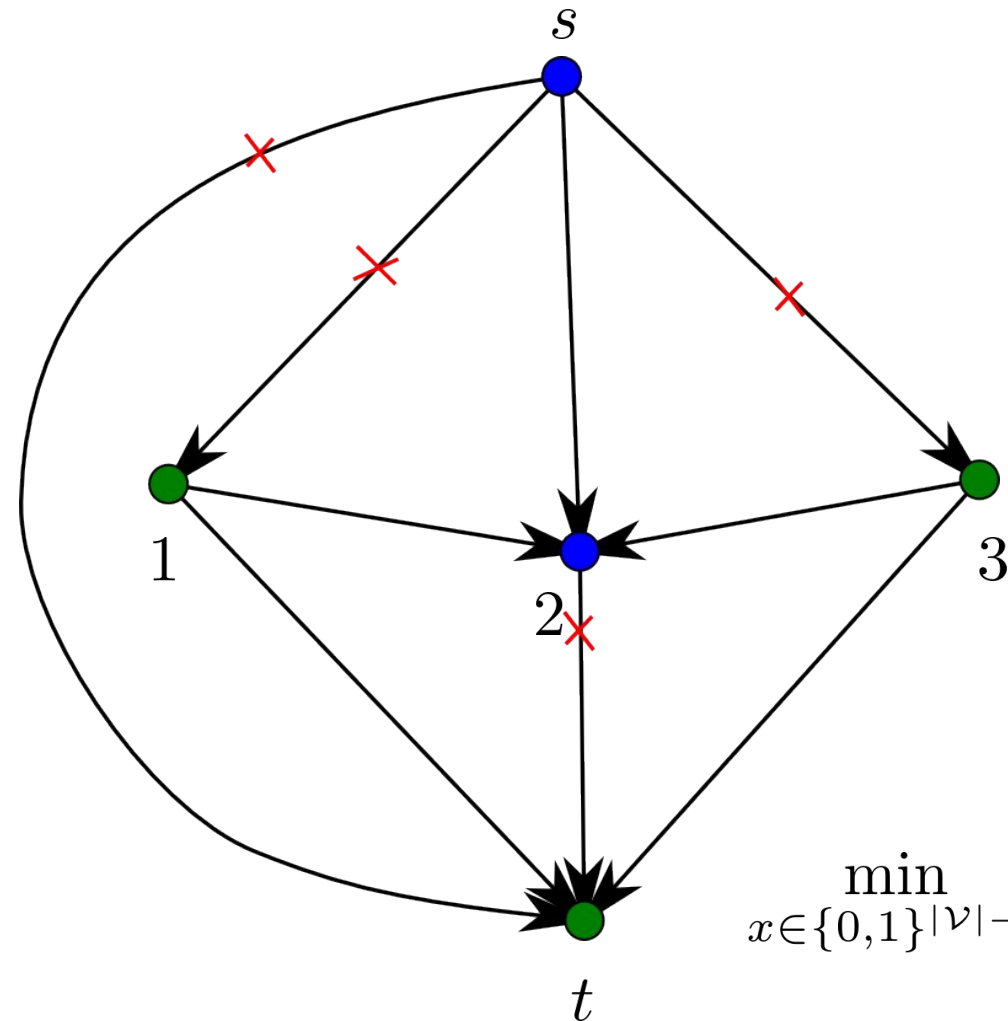
$$(x_i = 1) \Rightarrow (i \in T)$$

$$u \xrightarrow{c_{uv} \bar{x}_u x_v} v$$

$$s \xrightarrow{c_{sv} x_v} v$$

$$u \xrightarrow{c_{ut} \bar{x}_u} t$$

Cut as Oriented Quadratic Pseudo-Boolean Function



st -Min-cut problem:

$$\min_{S,T} \sum_{\substack{u \in S, v \in T \\ uv \in \mathcal{E}'}} c_{uv}$$

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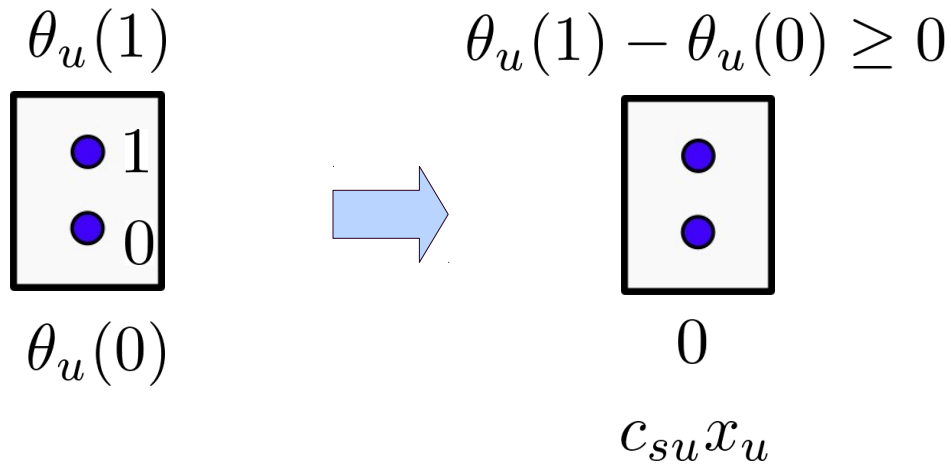
$$\min_{x \in \{0,1\}^{|\mathcal{V}|-2}} c_{st} + c_{s1}x_1 + c_{s2}x_2 + c_{s3}x_3$$

$$+ c_{12}\bar{x}_1x_2 + c_{32}\bar{x}_3x_2$$

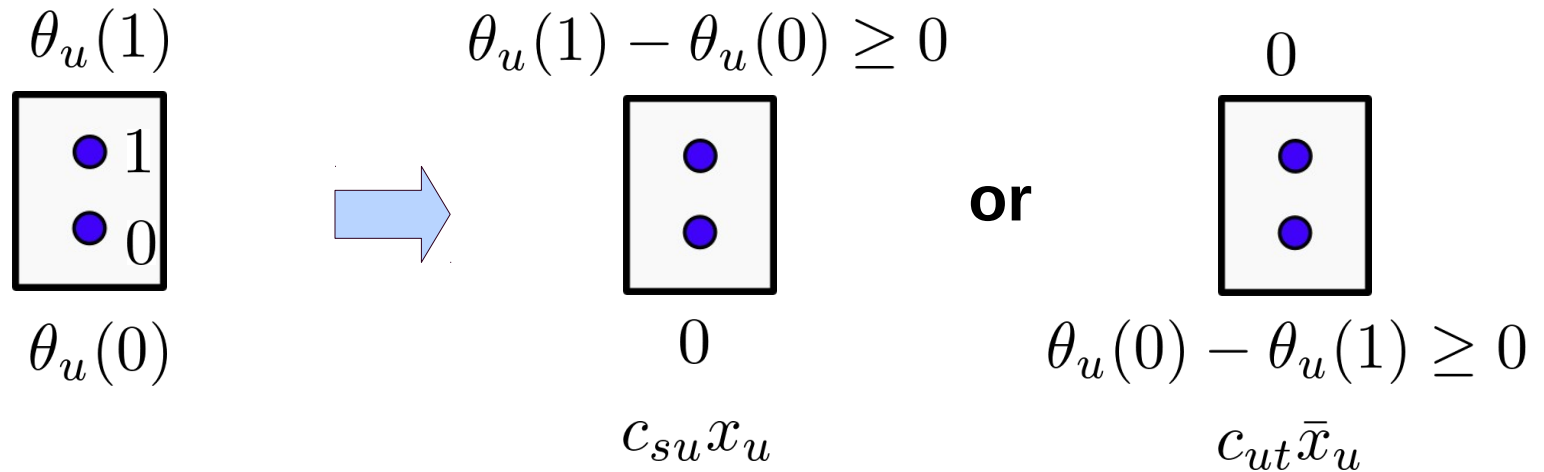
$$+ c_{1t}\bar{x}_1 + c_{3t}\bar{x}_3 + c_{3t}\bar{x}_3$$

$$\begin{array}{c} \theta_u(1) \\ \boxed{\begin{array}{c} \bullet 1 \\ \bullet 0 \end{array}} \\ \theta_u(0) \end{array}$$

Binary Energy as Oriented Quadratic Pseudo-Boolean Function

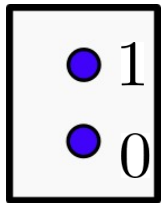


Binary Energy as Oriented Quadratic Pseudo-Boolean Function

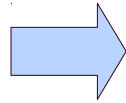


Binary Energy as Oriented Quadratic Pseudo-Boolean Function

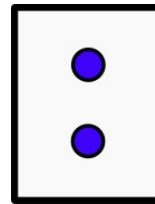
$$\theta_u(1)$$



$$\theta_u(0)$$



$$\theta_u(1) - \theta_u(0) \geq 0$$

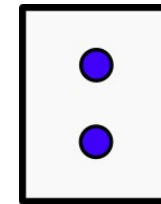


$$0$$

$$c_{su}x_u$$

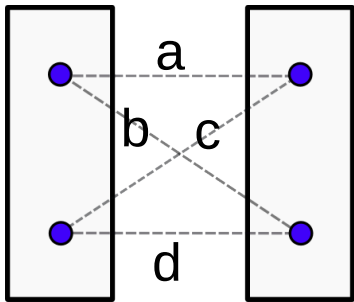
or

$$0$$

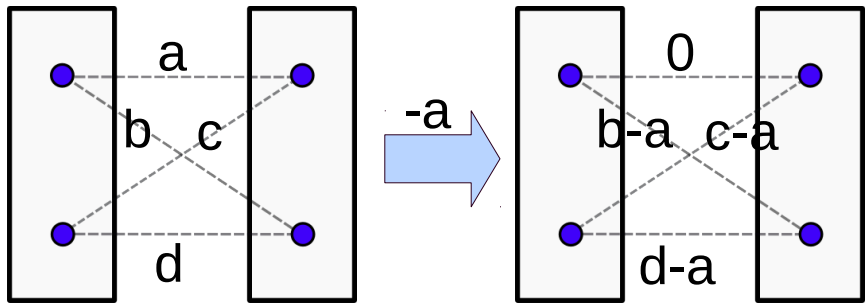
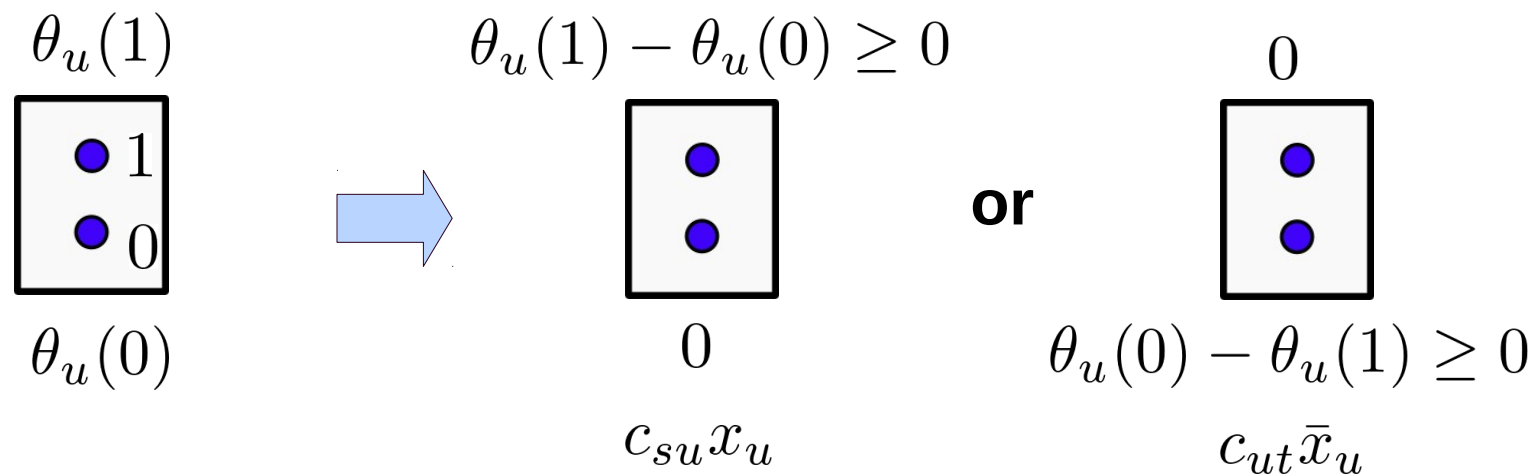


$$\theta_u(0) - \theta_u(1) \geq 0$$

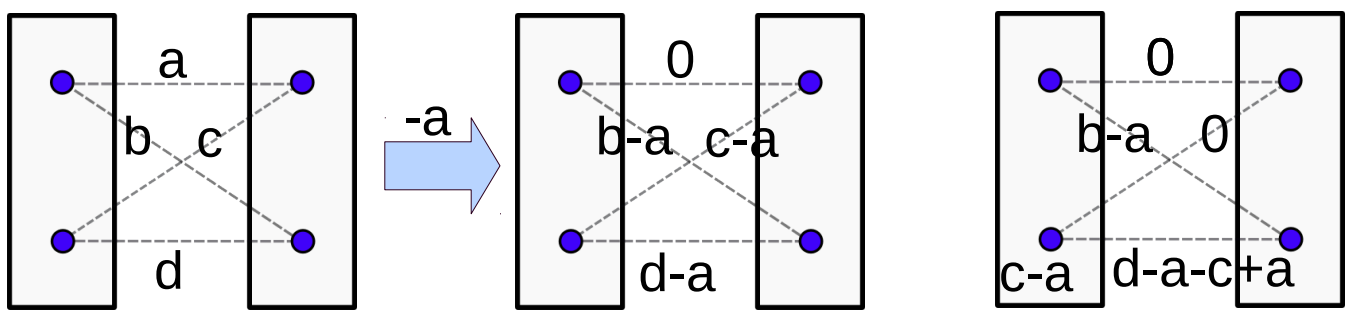
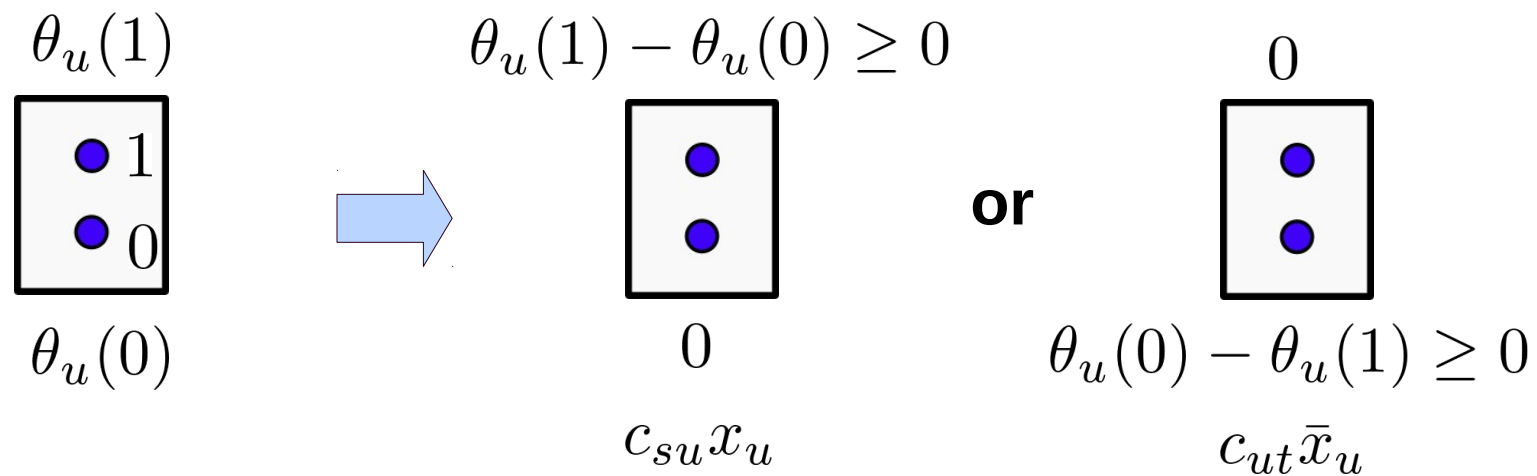
$$c_{ut}\bar{x}_u$$



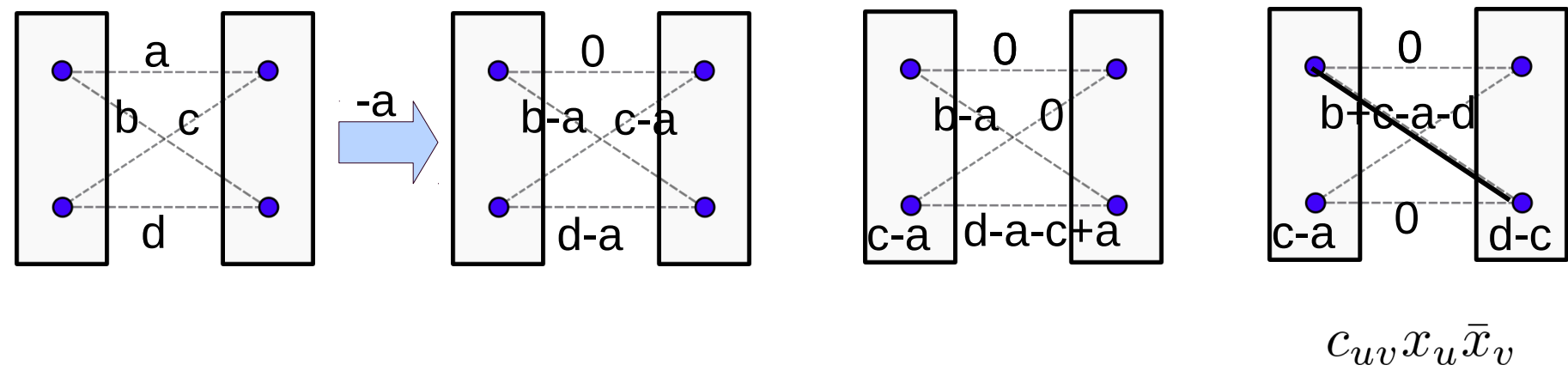
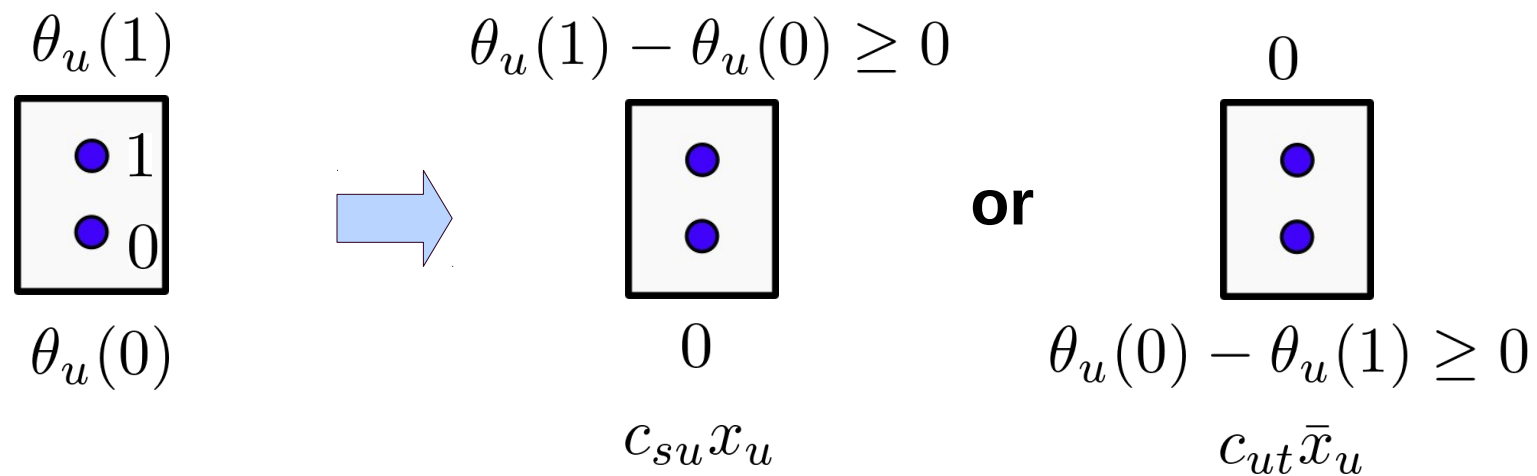
Binary Energy as Oriented Quadratic Pseudo-Boolean Function



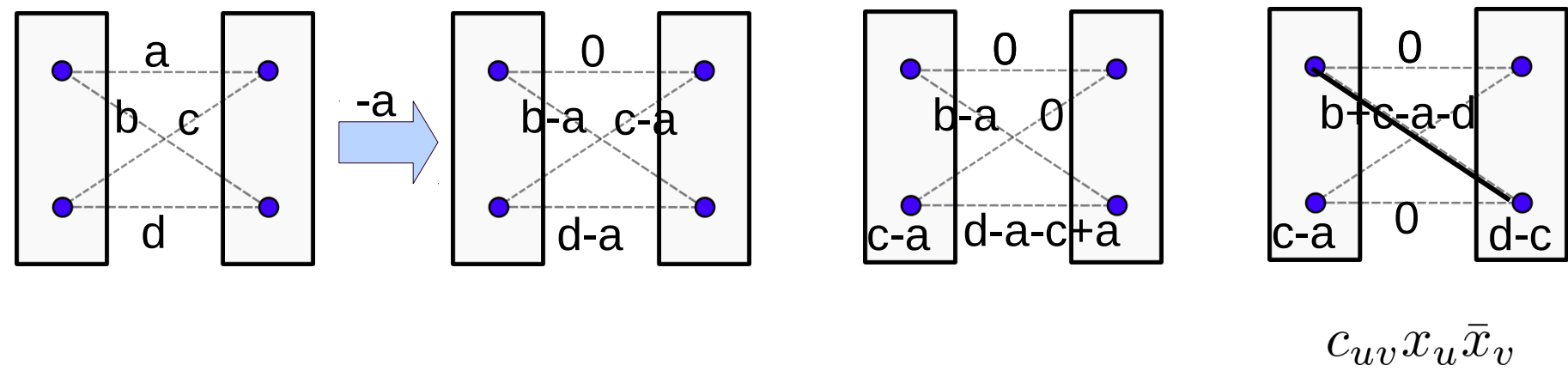
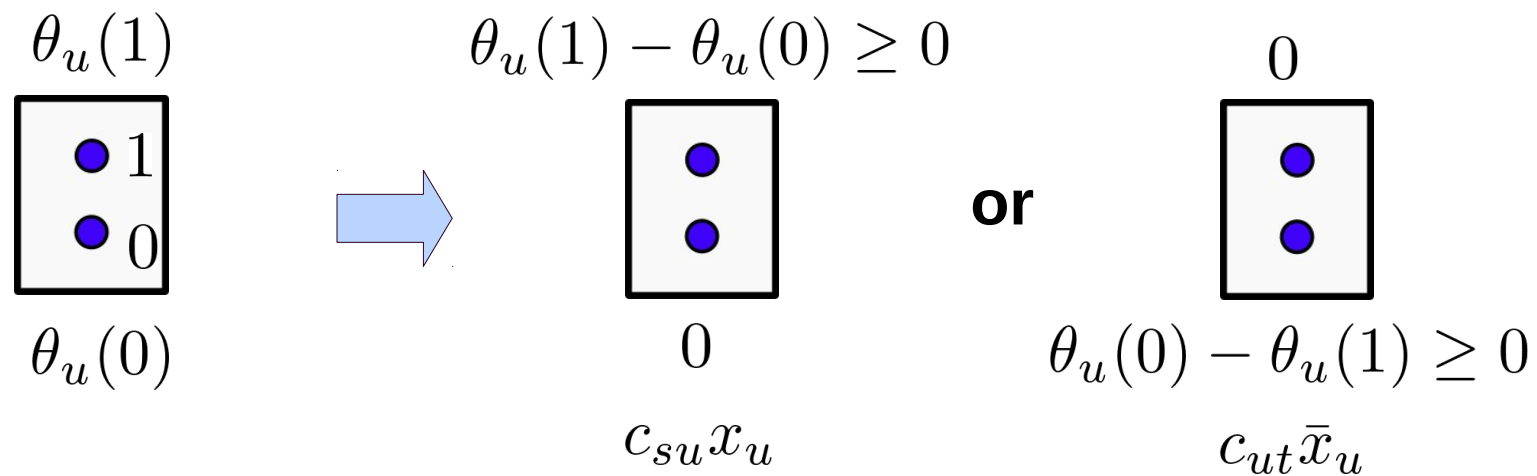
Binary Energy as Oriented Quadratic Pseudo-Boolean Function



Binary Energy as Oriented Quadratic Pseudo-Boolean Function

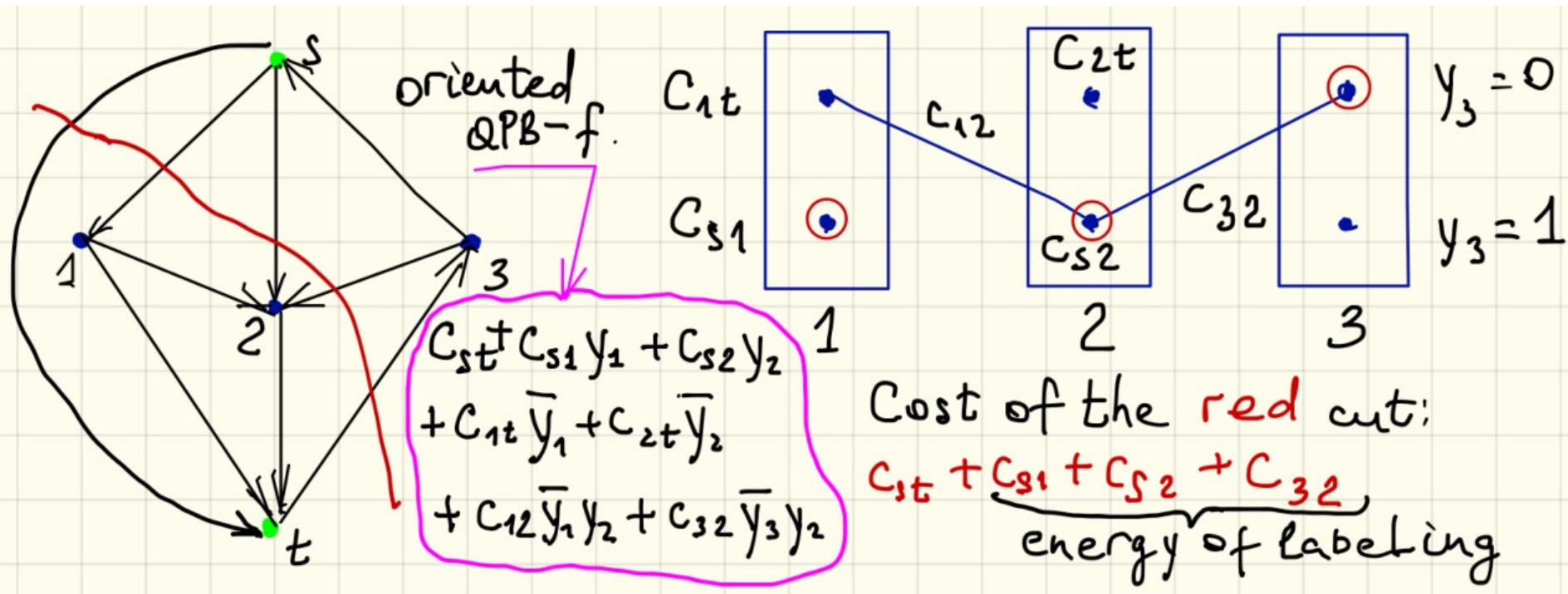


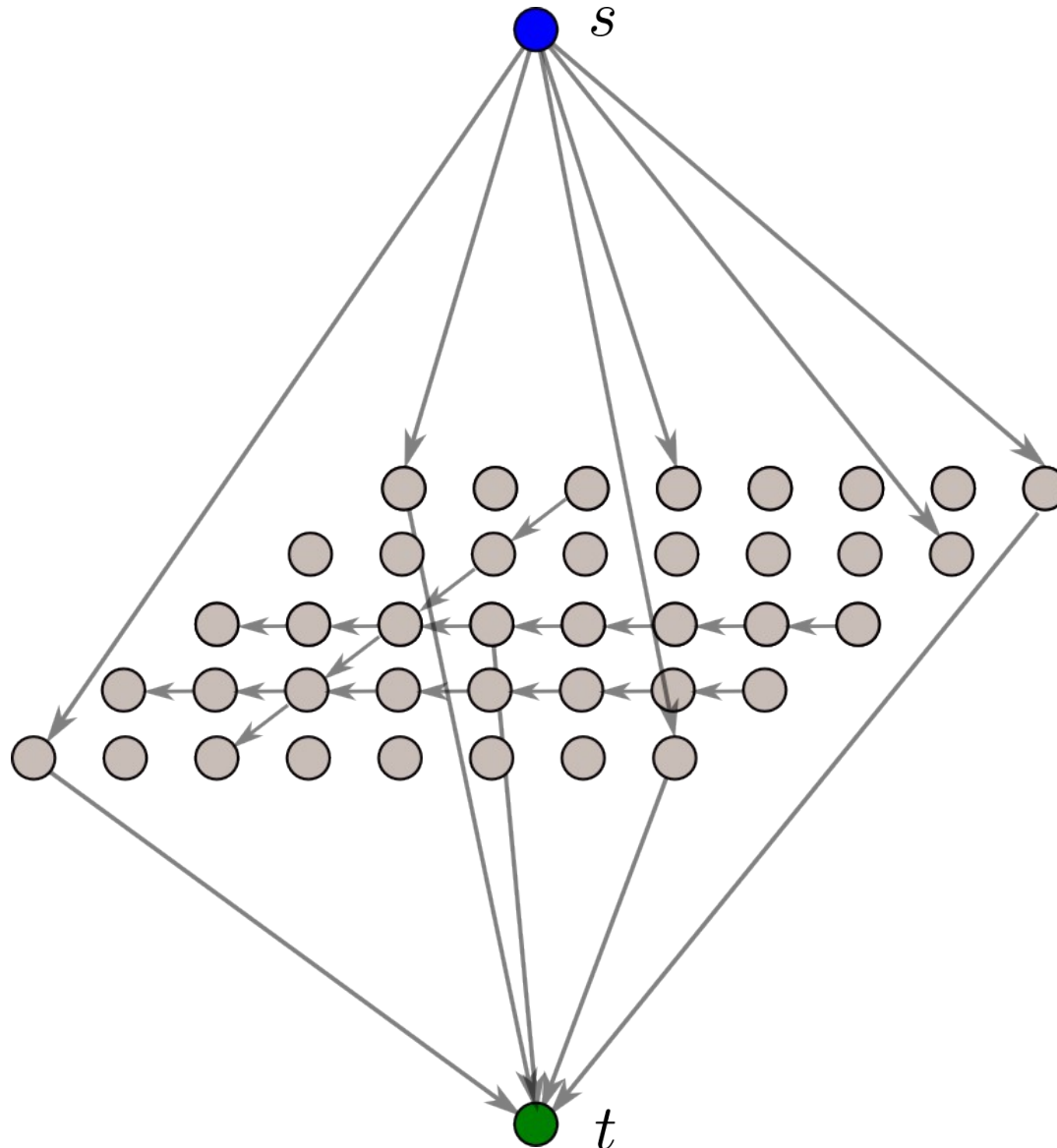
Binary Energy as Oriented Quadratic Pseudo-Boolean Function



If θ_{uv} submodular $\Rightarrow c_{uv} \geq 0$

Binary Energy as Min-Cut





$$\text{CUT} \Leftrightarrow c_{su}x_u + c_{uv}\bar{x}_ux_v + c_{ut}\bar{x}_u \Leftrightarrow \text{Binary energy}$$

$$\text{CUT} \Leftrightarrow c_{su}x_u + c_{uv}\bar{x}_ux_v + c_{ut}\bar{x}_u \Leftrightarrow \text{Binary energy}$$

$$\text{MIN-CUT} \Leftrightarrow \min_x c_{su}x_u + c_{uv}\bar{x}_ux_v + c_{ut}\bar{x}_u \Leftrightarrow \min_x (\text{binary energy})$$

$$\text{CUT} \Leftrightarrow c_{su}x_u + c_{uv}\bar{x}_ux_v + c_{ut}\bar{x}_u \Leftrightarrow \text{Binary energy}$$

$$\text{MIN-CUT} \Leftrightarrow \min_x c_{su}x_u + c_{uv}\bar{x}_ux_v + c_{ut}\bar{x}_u \Leftrightarrow \min_x (\text{binary energy})$$

$$\text{solvable MIN-CUT} \Leftrightarrow c_{uv} \geq 0 \Leftrightarrow \min_x (\text{submodular binary energy})$$

Your conclusions are welcome!